



Electrodynamics

电动力学

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容与

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1 Maxwell 电磁场方程

1.1 Maxwell 方程组的导出

1.1.1 电荷守恒定律

Law. 1 电荷守恒定律

$$\nabla \cdot \mathbf{J}(t, \mathbf{x}) + \frac{\partial \rho(t, \mathbf{x})}{\partial t} = 0 \quad (1.1)$$

Cor. 1 稳恒电流的散度

对于稳恒体系，物理量不显含时间， $\frac{\partial \rho}{\partial t} = 0$ ，从而

$$\nabla \cdot \mathbf{J} = 0 \quad (1.2)$$

Prop. 1 稳恒电流的边值关系

$$\mathbf{n} \cdot (\mathbf{J}_2 - \mathbf{J}_1) = 0 \quad \Longleftrightarrow \quad J_{2n} = J_{1n} \quad (1.3)$$

Thm. 1 守恒律普遍表达式

$$[\text{流密度的散度}] + [\text{空间密度的时间变化率}] = 0$$

称为连续性方程或流守恒方程.

1.1.2 静电场

Law. 2 Coulomb 定律

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qq'}{r^3} \mathbf{r} \quad (1.4)$$

其中 $\mathbf{r} = \mathbf{x} - \mathbf{x}'$, $r = |\mathbf{x} - \mathbf{x}'|$.

Def. 1 电场强度

$$\mathbf{E}(\mathbf{x}) \stackrel{\text{def}}{=} \frac{\mathbf{F}(\mathbf{x})}{q} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{x}')\mathbf{r}}{r^3} d^3x' \quad (1.5)$$

Lem. 1

$$\nabla \frac{1}{r} = -\frac{\mathbf{r}}{r^3} \quad (1.6)$$

$$\nabla^2 \frac{1}{r} = -\nabla \cdot \frac{\mathbf{r}}{r^3} = -4\pi\delta(\mathbf{r}) \quad (1.7)$$

$$\nabla \times \nabla\phi = 0 \quad (1.8)$$

Thm. 2 静电场的散度: 电场的 Gauss 定理

$$\nabla \cdot \mathbf{E}(\mathbf{x}) = \frac{\rho(\mathbf{x})}{\varepsilon_0} \quad (1.9)$$

$$\begin{aligned} \text{Pf } \nabla \cdot \mathbf{E}(\mathbf{x}) &= \nabla \cdot \left(\frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{x}')\mathbf{r}}{r^3} d^3x' \right) = \frac{1}{4\pi\varepsilon_0} \int \rho(\mathbf{x}') d^3x' \underbrace{\nabla \cdot \left(\frac{\mathbf{r}}{r^3} \right)}_{=4\pi\delta(\mathbf{r})}, \\ \Rightarrow \nabla \cdot \mathbf{E}(\mathbf{x}) &= \frac{1}{\varepsilon_0} \int \rho(\mathbf{x}')\delta(\mathbf{r}) d^3x' = \frac{\rho(\mathbf{x})}{\varepsilon_0}. \end{aligned}$$

Cor. 2 时变电场的散度

由于未使用恒定条件, 可推广为

$$\nabla \cdot \mathbf{E}(t, \mathbf{x}) = \frac{\rho(t, \mathbf{x})}{\varepsilon_0} \quad (1.10)$$

Thm. 3 静电场的旋度

$$\nabla \times \mathbf{E}(\mathbf{x}) = 0 \quad (1.11)$$

$$\begin{aligned} \text{Pf } \mathbf{E}(\mathbf{x}) &= \frac{1}{4\pi\varepsilon_0} \int \rho(\mathbf{x}') d^3x' \frac{\mathbf{r}}{r^3} = -\frac{1}{4\pi\varepsilon_0} \int \rho(\mathbf{x}') d^3x' \nabla \frac{1}{r} = -\nabla \underbrace{\left(\frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{x}')}{r} d^3x' \right)}_{\stackrel{\text{def}}{=} \phi(\mathbf{x})}, \\ \Rightarrow \nabla \times \mathbf{E}(\mathbf{x}) &= -\nabla \times (\nabla\phi) = 0. \end{aligned}$$

Def. 2 电势

$$\phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{x}')}{r} d^3x' \quad (1.12)$$

1.1.3 静磁场
Law. 3 Biot-Savart 定律

$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}') \times \mathbf{r}}{r^3} d^3x' = \frac{\mu_0}{4\pi} \oint \frac{I d\mathbf{x} \times \mathbf{r}}{r^3} \quad (1.13)$$

Lem. 2

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0 \quad (1.14)$$

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \quad (1.15)$$

Thm. 4 静磁场的散度

$$\nabla \cdot \mathbf{B}(\mathbf{x}) = 0 \quad (1.16)$$

$$\begin{aligned} \text{Pf } \frac{\mathbf{J} \times \mathbf{r}}{r^3} &= -\mathbf{J} \times \nabla \frac{1}{r} = \nabla \times \frac{\mathbf{J}}{r} - \underbrace{\frac{\nabla \times \mathbf{J}(\mathbf{x}')}{r}}_{=0}, \\ \Rightarrow \mathbf{B}(\mathbf{x}) &= \frac{\mu_0}{4\pi} \int \nabla \times \frac{\mathbf{J}(\mathbf{x}')}{r} d^3x' = \nabla \times \underbrace{\left(\frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}')}{r} d^3x' \right)}_{\stackrel{\text{def}}{=} \mathbf{A}}, \Rightarrow \nabla \cdot \mathbf{B}(\mathbf{x}) = \nabla \cdot (\nabla \times \mathbf{A}) = 0. \end{aligned}$$

Cor. 3 非恒定情况

$$\nabla \cdot \mathbf{B}(t, \mathbf{x}) = 0 \quad (1.17)$$

Def. 3 磁矢势

$$\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}')}{r} d^3x' \quad (1.18)$$

Thm. 5 静磁场的旋度: Ampère 环路定理

$$\nabla \times \mathbf{B}(\mathbf{x}) = \mu_0 \mathbf{J}(\mathbf{x}) \quad (1.19)$$

$$\begin{aligned} \text{Pf } \nabla \cdot \mathbf{A}(\mathbf{x}) &= \frac{\mu_0}{4\pi} \int \nabla \cdot \frac{\mathbf{J}(\mathbf{x}')}{r} d^3x'. \quad \nabla \cdot \frac{\mathbf{J}(\mathbf{x}')}{r} = \nabla \cdot \frac{1}{r} \cdot \mathbf{J} = -\nabla' \cdot \frac{1}{r} \cdot \mathbf{J} = \frac{\nabla' \cdot \mathbf{J}}{r} - \nabla' \cdot \frac{\mathbf{J}}{r}. \\ \text{由电流恒定条件, 有 } \nabla' \cdot \mathbf{J}(\mathbf{x}') &= 0, \int \nabla' \cdot \frac{\mathbf{J}}{r} d^3x' = \oint \frac{\mathbf{J}}{r} \cdot \mathbf{n} d^2x' = 0, \Rightarrow \nabla \cdot \mathbf{A}(\mathbf{x}) = 0. \\ &\quad \text{Coulomb 规范} \\ \nabla^2 \mathbf{A}(\mathbf{x}) &= \frac{\mu_0}{4\pi} \int \nabla^2 \frac{1}{r} \mathbf{J}(\mathbf{x}') d^3x' = -\mu_0 \int \delta(\mathbf{r}) \mathbf{J}(\mathbf{x}') d^3x' = \mu_0 \mathbf{J}(\mathbf{x}). \Rightarrow \nabla \times \mathbf{B} = \mu_0 \mathbf{J}(\mathbf{x}). \end{aligned}$$

Lem. 3 柱坐标中的旋度

$$\nabla \times \mathbf{B} = \frac{1}{r} \begin{vmatrix} \mathbf{e}_r & r\mathbf{e}_\theta & \mathbf{e}_z \\ \partial_r & \partial_\theta & \partial_z \\ B_r & rB_\theta & B_z \end{vmatrix} \quad (1.20)$$

1.1.4 感生电场

Law. 4 Faraday 电磁感应定律

$$\mathcal{E} = -\frac{d\Phi_B}{dt} \quad (1.21)$$

Thm. 6 感生电场的旋度

$$\nabla \times \mathbf{E}(t, \mathbf{x}) = -\frac{\partial \mathbf{B}(t, \mathbf{x})}{\partial t} \quad (1.22)$$

Rmk 变化的磁场激发感生涡旋电场.

$$\text{Pf } \mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{x} = -\frac{d\Phi_B}{dt} = \int (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = -\frac{d}{dt} \int \mathbf{B} \cdot d\mathbf{S} = -\int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}.$$

1.1.5 位移电流

Def. 4 位移电流

$$\mathbf{J}_D = \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (1.23)$$

Rmk 变化的电场激发磁场.

$$\text{Pf } 0 = \nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 \left(\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} \right) = \mu_0 \left(\nabla \cdot \mathbf{J} + \varepsilon_0 \frac{\partial}{\partial t} \nabla \cdot \mathbf{E} \right) = \mu_0 \nabla \cdot (\mathbf{J} + \mathbf{J}_D).$$

Thm. 7 磁场的旋度: Maxwell-Ampère 定理

$$\nabla \times \mathbf{B}(t, \mathbf{x}) = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (1.24)$$

1.1.6 真空中 Maxwell 方程组

Thm. 8 真空中 Maxwell 方程组

$$\begin{cases} \nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \end{cases} \quad (1.25)$$

1.2 介质中的电磁场方程

介质对外场的响应

外场的作用使介质分子的电偶极矩与分子电流的磁偶极矩取向呈现规律性, 导致出现宏观的电荷与电流分布, 其激发的电磁场与外场叠加即为总场.

$$\begin{cases} \rho_{\text{total}} = \rho + \rho_{\text{P}} \\ \mathbf{J}_{\text{total}} = \mathbf{J} + \mathbf{J}_{\text{P}} + \mathbf{J}_{\text{M}} \end{cases} \quad (1.26)$$

其中 $\rho, \mathbf{J}$ 分别为自由电荷密度与自由电流密度. 自由电荷面密度则为 σ , 线密度为 λ .

1.2.1 介质的极化

Def. 5 极化强度

$$\mathbf{P} = \lim_{\Delta V \rightarrow 0} \frac{1}{\Delta V} \sum_i \mathbf{p}_i = n\mathbf{p} \quad (1.27)$$

Prop. 2 极化电荷的体分布

$$\rho_{\text{P}} = -\nabla \cdot \mathbf{P} \quad (1.28)$$

Rmk 均匀极化的介质内, 极化电荷只分布于自由电荷附近与介质界面处.

$$\text{Pf} \quad \int \rho_{\text{P}}(\mathbf{x}) d^3x = -\oint nql \cdot d^2\mathbf{x} = -\oint \underbrace{\mathbf{P} \cdot \mathbf{n}}_{\stackrel{\text{def}}{=} \sigma_{\text{P}}} d^2x = -\int \nabla \cdot \mathbf{P} d^3x.$$

Prop. 3 极化电荷的面分布

$$\sigma_{\text{P}} = -\mathbf{n} \cdot (\mathbf{P}_2 - \mathbf{P}_1) = -(P_{2n} - P_{1n}) = \varepsilon_0(E_{2n} - E_{1n}) - \sigma \quad (1.29)$$

其中 $\mathbf{n}$ 为从介质 1 指向介质 2 的单位法矢量.

$$\text{Pf} \quad \sigma_{\text{P}} \Delta S = -\oint \mathbf{n} \cdot \mathbf{P} d^2x = -\mathbf{n} \cdot (\mathbf{P}_2 - \mathbf{P}_1) \Delta S.$$

Prop. 4 极化电流

$$\mathbf{J}_{\text{P}} = \frac{\partial \mathbf{P}}{\partial t} \quad (1.30)$$

$$\text{Pf} \quad \text{由电荷守恒律, } \nabla \cdot \mathbf{J}_{\text{P}} = -\frac{\partial \rho_{\text{P}}}{\partial t} = \nabla \cdot \frac{\partial \mathbf{P}}{\partial t}.$$

Cor. 4 均匀极化体的总极矩

若介质体均匀极化, 即极化强度为常矢量 $\mathbf{P}$, 其体积为 V , 则其总极矩为

$$\mathbf{p}_{\text{total}} = \mathbf{P}V \quad (1.31)$$

Lem. 4 二阶张量的 Gauss 公式

$$\int_{\Omega} \nabla \cdot \mathcal{T} \, d^3x = \oint_{\partial\Omega} \mathbf{n} \cdot \mathcal{T} \, d^2x \quad (1.32)$$

Prop. 5 孤立介质所受电场力

$$\mathbf{F} = \int \rho \mathbf{E}_0 + \mathbf{P} \cdot \nabla \mathbf{E}_0 \, d^3x \quad (1.33)$$

$$\begin{aligned} \text{Pf } \mathbf{F} &= \int (\rho + \rho_P) \mathbf{E}_0 \, d^3x + \oint \sigma_P \mathbf{E}_0 \, d^2x = \int (\rho - \nabla \cdot \mathbf{P}) \mathbf{E}_0 \, d^3x + \oint \mathbf{n} \cdot \mathbf{P} \mathbf{E} \, d^2x \\ &= \int (\rho - \nabla \cdot \mathbf{P}) \mathbf{E}_0 + \nabla \cdot (\mathbf{P} \mathbf{E}_0) \, d^3x = \int \rho \mathbf{E}_0 \underbrace{-(\nabla \cdot \mathbf{P}) \mathbf{E}_0 + (\nabla \cdot \mathbf{P}) \mathbf{E}_0}_{=0} + \mathbf{P} \cdot \nabla \mathbf{E}_0 \, d^3x. \end{aligned}$$

1.2.2 介质的磁化**Def. 6 磁化强度**

$$\mathbf{M} = \lim_{\Delta V \rightarrow 0} \frac{1}{\Delta V} \sum_i \mathbf{M}_i = n\mathbf{m} \quad (1.34)$$

Prop. 6 磁化电流的体分布

$$\mathbf{J}_M = \nabla \times \mathbf{M} \quad (1.35)$$

$$\text{Rmk } \rho_M = \nabla \cdot \mathbf{J}_M = \nabla \cdot (\nabla \times \mathbf{M}) \equiv 0.$$

$$\text{Pf } I_M = \int \mathbf{J}_M \cdot d^2\mathbf{x} = \oint \underbrace{n\mathbf{i}\mathbf{S}}_{=n\mathbf{m}} \cdot d\mathbf{x} = \oint \mathbf{M} \cdot d\mathbf{x} = \int \nabla \times \mathbf{M} \cdot d^2\mathbf{x}.$$

Prop. 7 磁化电流的面分布

$$\mathbf{K}_M = \mathbf{n} \times (\mathbf{M}_2 - \mathbf{M}_1) = \frac{1}{\mu_0} \mathbf{n} \times (\mathbf{B}_2 - \mathbf{B}_1) - \mathbf{K} \quad (1.36)$$

Prop. 8 磁化电流的线分布

$$I_M = \oint \mathbf{M} \cdot d\mathbf{x} \quad (1.37)$$

Cor. 5 均匀磁化体的总磁矩

若介质体均匀磁化, 即磁化强度为常矢量 $\mathbf{M}$, 其体积为 V , 则其总磁矩为

$$\mathbf{m}_{\text{total}} = \mathbf{M}V \quad (1.38)$$

1.2.3 介质中的 Maxwell 方程组**Def. 7 电位移矢量**

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} \quad (1.39)$$

$$\text{Pf } \nabla \cdot \mathbf{E} = \frac{\rho_{\text{total}}}{\varepsilon_0}, \Rightarrow \nabla \cdot \underbrace{\varepsilon_0 \mathbf{E} + \mathbf{P}}_{\text{def}} = \rho.$$

Thm. 9 电场关系式

对于各向同性的均匀线性介质, 假设介质对场的响应是瞬时定域的, 则 $\mathbf{P}$ 与 $\mathbf{E}$ 有线性关系 $\mathbf{P} = \chi_e \varepsilon_0 \mathbf{E}$, 其中 χ_e 称为介质的极化率.

设 $\varepsilon = \varepsilon_0(1 + \chi_e) = \varepsilon_0 \varepsilon_r$, 则 $\mathbf{D}$ 与 $\mathbf{E}$ 有线性关系

$$\mathbf{D} = \varepsilon \mathbf{E} \quad (1.40)$$

其中 ε 称为介电常量, 或电容率; ε_r 称为相对介电常量, 或相对电容率.

1. 对于非均匀介质, 其介电常量为场点位置的函数

$$\mathbf{D}(\mathbf{x}) = \varepsilon(\mathbf{x}) \mathbf{E}(\mathbf{x}) \quad (1.41)$$

2. 对于各项异性介质, 由介电常量张量 ε_{ab} 描述介质对外场的响应

$$D_a = \varepsilon_{ab} E^b \quad (1.42)$$

3. 对于非线性介质, $\mathbf{D}$ 对 $\mathbf{E}$ 非线性依赖

$$D_a = \varepsilon_{ab} E^b + \varepsilon_{abc} E^b E^c + \varepsilon_{abcd} E^b E^c E^d + \dots \quad (1.43)$$

Def. 8 磁场强度

$$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \quad (1.44)$$

即 $\mathbf{B} = \mu_0 \mathbf{H} + \mu_0 \mathbf{M}$. 设 $\mathbf{M} = \chi_m \mathbf{H}$, $\mu = \mu_0(1 + \chi_m)$, 则

$$\mathbf{B} = \mu \mathbf{H} \quad (1.45)$$

Rmk 非均匀、各项异性、非线性介质的磁场强度, 可与电位移矢量类似地定义.

Thm. 10 介质中的 Maxwell 方程组

$$\begin{cases} \nabla \cdot \mathbf{D} = \rho \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \end{cases} \quad (1.46)$$

$$\text{Pf } \nabla \cdot \mathbf{E} = \frac{\rho + \rho_P}{\varepsilon_0} = \frac{\rho - \nabla \cdot \mathbf{P}}{\varepsilon_0}, \Rightarrow \nabla \cdot (\varepsilon_0 \mathbf{E} + \mathbf{P}) = \nabla \cdot \mathbf{D} = \rho.$$

$$\nabla \times \mathbf{B} = \mu_0(\mathbf{J} + \mathbf{J}_P + \mathbf{J}_M) + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mu_0(\mathbf{J} + \nabla \times \mathbf{M}) + \mu_0 \frac{\partial}{\partial t} (\underbrace{\varepsilon_0 \mathbf{E} + \mathbf{P}}_{=\mathbf{D}}),$$

$$\Rightarrow \nabla \times (\underbrace{\mathbf{B}/\mu_0 - \mathbf{M}}_{=\mathbf{H}}) = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}.$$

Thm. 11 本构方程

$$\begin{cases} \mathbf{D} = \varepsilon \mathbf{E} \\ \mathbf{H} = \frac{1}{\mu} \mathbf{B} \\ \mathbf{J} = \sigma \mathbf{E} \end{cases} \quad (1.47)$$

1.3 电磁场的边值关系

Lem. 5 积分形式的 Maxwell 方程组

$$\begin{cases} \oint_{\partial\Omega} \mathbf{D} \cdot d^2\mathbf{x} = \int_{\Omega} \rho d^3x \\ \oint_{\partial\Sigma} \mathbf{E} \cdot d\mathbf{x} = -\frac{d}{dt} \int_{\Sigma} \mathbf{B}_{\Sigma} \cdot d^2\mathbf{x} \\ \oint_{\partial\Omega} \mathbf{B} \cdot d^2\mathbf{x} = 0 \\ \oint_{\partial\Sigma} \mathbf{H} \cdot d\mathbf{x} = \int_{\Sigma} \mathbf{J} \cdot d^2\mathbf{x} + \frac{d}{dt} \int_{\Sigma} \mathbf{D} \cdot d^2\mathbf{x} \end{cases} \quad (1.48)$$

Thm. 12 介质分界面两侧的边值关系

设 $\mathbf{n}$ 为介质 1 指向介质 2 的单位法矢量,

$$\begin{cases} \mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \sigma \\ \mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) = 0 \\ \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0 \\ \mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{K} \end{cases} \quad (1.49)$$

或可写为

$$\begin{cases} D_{2n} - D_{1n} = \sigma \\ E_{2t} = E_{1t} \\ B_{2n} = B_{1n} \\ H_{2t} - H_{1t} = K \end{cases} \quad (1.50)$$

1.4 Maxwell 方程组的对称性 ***1.4.1 时空对称性****1.4.2 Lorentz 协变性****1.4.3 规范对称性****Def. 9 Lorenz 规范**

$$\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} = 0 \quad (1.51)$$

Def. 10 Coulomb 规范

$$\nabla \cdot \mathbf{A} = 0 \quad (1.52)$$

1.4.4 分立对称性**1.5 Lorentz 力****Def. 11 Lorentz 力密度**

$$\mathbf{f} = \rho \mathbf{E} + \mathbf{J} \times \mathbf{B} \quad (1.53)$$

Def. 12 Lorentz 力

带电荷量 q , 以速度 $\boldsymbol{v}$ 运动的带电粒子在电磁场中所受力为

$$\boldsymbol{F} = q\boldsymbol{E} + q\boldsymbol{v} \times \boldsymbol{B} \tag{1.54}$$

2 电磁场的守恒律

2.1 电磁场能量

2.1.1 能量守恒定律

Law. 1 开放体系的能量守恒定律

$$\oint_{\partial\Omega} \mathbf{S} \cdot d^2\mathbf{x} + \frac{d}{dt} \int_{\Omega} w d^3x + \int_{\Omega} \mathbf{f} \cdot \mathbf{v} d^3x = 0 \quad (2.1)$$

Thm. 1 能量守恒律的微分形式

$$\nabla \cdot \mathbf{S} + \frac{\partial w}{\partial t} + \mathbf{f} \cdot \mathbf{v} = 0 \quad (2.2)$$

Cor. 1 孤立系统或全空间的能量守恒律

对于封闭体系 Ω 或全空间, 边界面上无能量流动, 即

$$\oint_{\partial\Omega/\infty} \mathbf{S} \cdot d^2\mathbf{x} = 0 \quad (2.3)$$

电磁场能量与带电体的机械能可相互转化

$$\int_{\Omega/\infty} \mathbf{f} \cdot \mathbf{v} d^3x = -\frac{d}{dt} \int_{\Omega/\infty} w d^3x \quad (2.4)$$

2.1.2 能量密度与能流密度

Prop. 1 能流密度与能量密度的表达式

对于一般介质, 能流密度 (Poynting 矢量) 与能量密度的时间变化率为

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (2.5)$$

$$\frac{\partial w}{\partial t} = \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} + \mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} \quad (2.6)$$

对于各向同性的非色散线性介质, 能量密度为

$$w = \frac{1}{2}(\mathbf{E} \cdot \mathbf{D} + \mathbf{H} \cdot \mathbf{B}) \quad (2.7)$$

另对于各向同性的均匀介质

$$\mathbf{S} = \frac{1}{\mu} \mathbf{E} \times \mathbf{B} \quad w = \frac{1}{2} \left(\epsilon E^2 + \frac{1}{\mu} B^2 \right) \quad (2.8)$$

$$\text{Pf} \quad \int \mathbf{f} \cdot \mathbf{v} d^3x = \int \mathbf{J} \cdot \mathbf{E} d^3x, \quad \mathbf{J} \cdot \mathbf{E} = \left(\nabla \times \mathbf{H} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot \mathbf{E} = \mathbf{E} \cdot (\nabla \times \mathbf{H}) - \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t}.$$

$$\mathbf{E} \cdot (\nabla \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{H}).$$

$$\text{若介质线性非色散, 则有 } \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} = \frac{\partial}{\partial t} \frac{\mathbf{E} \cdot \mathbf{D}}{2}, \quad \mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} = \frac{\partial}{\partial t} \frac{\mathbf{H} \cdot \mathbf{B}}{2}.$$

$$\Rightarrow \mathbf{f} \cdot \mathbf{v} = -\nabla \cdot \underbrace{(\mathbf{E} \times \mathbf{H})}_{\stackrel{\text{def}}{=} \mathbf{S}} - \frac{\partial}{\partial t} \underbrace{\frac{\mathbf{E} \cdot \mathbf{D} + \mathbf{H} \cdot \mathbf{B}}{2}}_{\stackrel{\text{def}}{=} w}$$

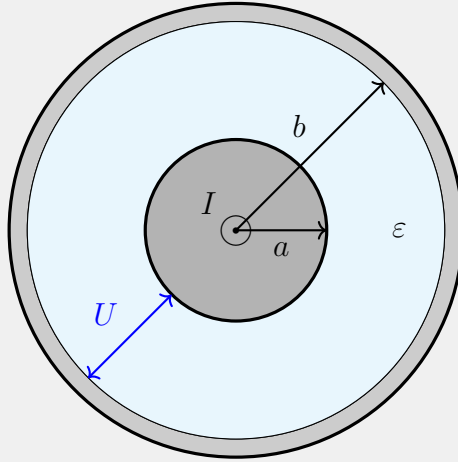
2.1.3 电磁能量的传输

Eg. 1 电容器的能量流动

Eg. 2 同轴电缆的能量流动

$$\mathbf{S} = \frac{UI}{2\pi r^2 \ln(b/a)} \mathbf{e}_z \quad (2.9)$$

$$S_r = -\frac{I^2}{2\pi^2 a^3 \sigma} \quad (2.10)$$



2.2 电磁场动量

2.2.1 动量守恒定律

Def. 1 Minkowski 力密度

$$\mathbf{f}_M = \rho \mathbf{E} + \mathbf{J} \times \mathbf{B} - \frac{1}{2} E^2 \nabla \varepsilon - \frac{1}{2} B^2 \nabla \mu \quad (2.11)$$

Thm. 2 微分形式的动量定理

$$\mathbf{f}_M + \frac{\partial \mathbf{g}_M}{\partial t} = -\nabla \cdot \mathcal{T} \quad (2.12)$$

Pf $\mathbf{f} = \rho \mathbf{E} + \mathbf{J} \times \mathbf{B} = (\nabla \cdot \mathbf{D})\mathbf{E} + (\nabla \times \mathbf{H}) \times \mathbf{B} - \frac{\partial \mathbf{D}}{\partial t} \times \mathbf{B}$. 由 $\nabla \cdot \mathbf{B} = 0$, $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$, 有 $\mathbf{f} = (\nabla \cdot \mathbf{D})\mathbf{E} + (\nabla \cdot \mathbf{B})\mathbf{H} - \frac{\partial}{\partial t}(\mathbf{D} \times \mathbf{B}) - \mathbf{B} \times (\nabla \times \mathbf{H}) - \mathbf{D} \times (\nabla \times \mathbf{E})$.

由 $\begin{cases} \mathbf{B} \times (\nabla \times \mathbf{H}) = (\nabla \mathbf{H}) \cdot \mathbf{B} - (\mathbf{B} \cdot \nabla)\mathbf{H} \\ \mathbf{D} \times (\nabla \times \mathbf{E}) = (\nabla \mathbf{E}) \cdot \mathbf{D} - (\mathbf{D} \cdot \nabla)\mathbf{E} \end{cases}$ 与 $\begin{cases} (\nabla \cdot \mathbf{D})\mathbf{E} + (\mathbf{D} \cdot \nabla)\mathbf{E} = \nabla \cdot (\mathbf{D}\mathbf{E}) \\ (\nabla \cdot \mathbf{B})\mathbf{H} + (\mathbf{B} \cdot \nabla)\mathbf{H} = \nabla \cdot (\mathbf{B}\mathbf{H}) \end{cases}$

$$\mathbf{f} = \nabla \cdot (\mathbf{D}\mathbf{E} + \mathbf{B}\mathbf{H}) - \frac{\partial}{\partial t}(\mathbf{D} \times \mathbf{B}) - (\nabla \mathbf{H}) \cdot \mathbf{B} - (\nabla \mathbf{E}) \cdot \mathbf{D}$$

由 $\nabla(\mathbf{D} \cdot \mathbf{E}) = (\nabla \mathbf{D}) \cdot \mathbf{E} + (\nabla \mathbf{E}) \cdot \mathbf{D}$, $\nabla \mathbf{D} \cdot \mathbf{E} = \nabla(\varepsilon \mathbf{E}) \cdot \mathbf{E} = E^2 \nabla \varepsilon + (\nabla \mathbf{E}) \cdot \mathbf{D}$, 有 $(\nabla \mathbf{E}) \cdot \mathbf{D} = \frac{1}{2} [\nabla(\mathbf{D} \cdot \mathbf{E}) - E^2 \nabla \varepsilon]$, 同理 $(\nabla \mathbf{H}) \cdot \mathbf{B} = \frac{1}{2} [\nabla(\mathbf{B} \cdot \mathbf{H}) - H^2 \nabla \mu]$.

又由 $\nabla \phi = \nabla \cdot (\phi \mathcal{I})$, 可得

$$\underbrace{\mathbf{f} - \frac{1}{2}E^2 \nabla \varepsilon - \frac{1}{2}H^2 \nabla \mu}_{\stackrel{\text{def}}{=} \mathbf{f}_M} = -\nabla \cdot \underbrace{\left[\frac{1}{2}(\mathbf{D} \cdot \mathbf{E} + \mathbf{B} \cdot \mathbf{H})\mathcal{I} - \mathbf{D}\mathbf{E} - \mathbf{B}\mathbf{H} \right]}_{\stackrel{\text{def}}{=} \mathcal{T}} - \frac{\partial}{\partial t} \underbrace{(\mathbf{D} \times \mathbf{B})}_{\stackrel{\text{def}}{=} \mathbf{g}_M}$$

Law. 2 开放体系的动量守恒定律

对于开放体系 Ω , 带电体系与电磁场的总动量变化率等于从 Ω 外通过边界 $\partial\Omega$ 流入的动量, 即

$$\frac{d\mathbf{G}_{\text{mech}}}{dt} + \frac{d\mathbf{G}_{\text{em}}}{dt} = \int_{\Omega} \mathbf{f}_M d^3x + \frac{d}{dt} \int_{\Omega} \mathbf{g}_M d^3x = - \oint_{\partial\Omega} \mathbf{n} \cdot \mathcal{T} d^2x \quad (2.13)$$

Cor. 2 封闭体系的动量守恒定律

对于封闭体系 $\bar{\Omega}$ 或全空间 ∞ , 带电体系的机械动量与电磁场动量守恒, 即

$$\frac{d}{dt} (\mathbf{G}_{\text{mech}} + \mathbf{G}_{\text{em}}) = \int_{\bar{\Omega}/\infty} \mathbf{f}_M d^3x + \frac{d}{dt} \int_{\bar{\Omega}/\infty} \mathbf{g}_M d^3x = 0 \quad (2.14)$$

$$\implies \mathbf{G}_{\text{mech}} + \mathbf{G}_{\text{em}} = \text{Const} \quad (2.15)$$

2.2.2 动量密度

Def. 2 动量密度

电磁场的动量密度矢量为

$$\mathbf{g}_M = \mathbf{D} \times \mathbf{B} \quad (2.16)$$

其体积分即为区域 Ω 内的电磁场动量

$$\mathbf{G}_{\text{em}} = \int \mathbf{g} \, d^3x \quad (2.17)$$

Prop. 2

动量密度与能流密度方向相同.

$$\mathbf{g}_M = \mu \varepsilon \mathbf{E} \times \mathbf{H} = \frac{\mu_r \varepsilon_r}{c^2} \mathbf{S} \quad (2.18)$$

2.2.3 动量流密度 (Maxwell 应力张量)

Def. 3 动量流密度

$$\mathcal{T} = -\mathbf{D}\mathbf{E} - \mathbf{B}\mathbf{H} + \frac{1}{2}(\mathbf{D} \cdot \mathbf{E} + \mathbf{B} \cdot \mathbf{H})\mathcal{I} \quad (2.19)$$

$$T_{ij} = -\varepsilon E_i E_j - \frac{1}{\mu} B_i B_j + \frac{1}{2} \left(\varepsilon \delta_{ij} E^2 + \frac{1}{\mu} \delta_{ij} B^2 \right) \quad (2.20)$$

Prop. 3 光压

2.3 电磁场角动量

2.3.1 角动量守恒定律

Lem. 1

$$\mathbf{x} \times (\nabla \cdot \mathcal{T}) = -\nabla \cdot (\mathcal{T} \times \mathbf{x}) \quad (2.21)$$

$$\begin{aligned} \text{Pf } \mathbf{x} \times (\nabla \cdot \mathcal{T}) &= \varepsilon_{ijk} x_i (\nabla \cdot \mathcal{T})_j \mathbf{e}_k = \varepsilon_{ijk} x_i \partial_l T_{lj} \mathbf{e}_k = \varepsilon_{ijk} \partial_l (x_i T_{lj}) \mathbf{e}_k - \underbrace{\varepsilon_{ijk} T_{lj} \delta_{il}}_{=0} \mathbf{e}_k \\ -\nabla \cdot (\mathcal{T} \times \mathbf{x}) &= -\nabla \cdot (T_{lj} \mathbf{e}_l \mathbf{e}_j \times x_i \mathbf{e}_i) = -\nabla \cdot (\mathbf{e}_l x_i T_{lj} \varepsilon_{jik} \mathbf{e}_k) = \varepsilon_{ijk} \partial_l (x_i T_{lj}) \mathbf{e}_k. \end{aligned}$$

Thm. 3 微分形式的角动量定理

$$\mathbf{x} \times \mathbf{f}_M + \frac{\partial \mathbf{l}_{\text{em}}}{\partial t} = -\nabla \cdot \mathcal{M}_{\text{em}} \quad (2.22)$$

Pf 由 $\mathbf{f}_M + \frac{\partial \mathbf{g}_M}{\partial t} = -\nabla \cdot \mathcal{T}$ 可得 $\mathbf{x} \times \mathbf{f}_M + \mathbf{x} \times \frac{\partial \mathbf{g}_M}{\partial t} = -\mathbf{x} \times (\nabla \cdot \mathcal{T})$.

$$\mathbf{x} \times \frac{\partial \mathbf{g}_M}{\partial t} = \frac{\partial}{\partial t}(\mathbf{x} \times \mathbf{g}_M) - \underbrace{\frac{d\mathbf{x}}{dt} \times \mathbf{g}_M}_{=0} \stackrel{\text{def}}{=} \frac{\partial \mathbf{l}_{em}}{\partial t}$$

由引理, $-\mathbf{x} \times (\nabla \cdot \mathcal{T}) = -\nabla \cdot (-\mathcal{T} \times \mathbf{x}) \stackrel{\text{def}}{=} -\nabla \cdot \mathcal{M}_{em}$.

Law. 3 开放体系的角动量守恒定律

$$\frac{d\mathbf{L}_{mech}}{dt} + \frac{d\mathbf{L}_{em}}{dt} = \int_{\Omega} \mathbf{x} \times \mathbf{f}_M d^3x + \frac{d}{dt} \int_{\Omega} \mathbf{l}_{em} d^3x = - \oint_{\partial\Omega} \mathbf{n} \cdot \mathcal{M}_{em} d^2x \quad (2.23)$$

Cor. 3 封闭体系的角动量守恒定律

对于封闭体系 $\bar{\Omega}$ 或全空间 ∞ , 带电体系的机械角动量与电磁场角动量守恒, 即

$$\frac{d}{dt} (\mathbf{L}_{mech} + \mathbf{L}_{em}) = \int_{\bar{\Omega}/\infty} \mathbf{x} \times \mathbf{f}_M d^3x + \frac{d}{dt} \int_{\bar{\Omega}/\infty} \mathbf{l}_{em} d^3x = 0 \quad (2.24)$$

$$\implies \mathbf{L}_{mech} + \mathbf{L}_{em} = \text{Const} \quad (2.25)$$

2.3.2 角动量密度

Def. 4 角动量密度

$$\mathbf{l}_{em} = \mathbf{x} \times \mathbf{g}_M \quad (2.26)$$

2.3.3 角动量流密度

Def. 5 角动量流密度

$$\mathcal{M}_{em} = -\mathcal{T} \times \mathbf{x} \quad (2.27)$$

3 静电学

Lem. 1 静电场基本方程

$$\begin{cases} \nabla \cdot \mathbf{D} = \rho \\ \nabla \times \mathbf{E} = 0 \\ \mathbf{D} = \varepsilon \mathbf{E} \end{cases} \quad (3.1)$$

$$\begin{cases} \mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \sigma \\ \mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) = 0 \end{cases}$$

3.1 静电势

3.1.1 静电势的微分方程

Prop. 1 电场强度的静电势表示

$$\mathbf{E} = -\nabla \phi \quad (3.2)$$

$$E_n = -\frac{\partial \phi}{\partial n} \quad (3.3)$$

$$d\phi = -\mathbf{E} \cdot d\mathbf{x} \quad (3.4)$$

Cor. 1 静电势的求解

设 $\phi(p_0) = 0$, 即取 p_0 为参考点, 则

$$\phi(p) = -\int_{p_0}^p \mathbf{E} \cdot d\mathbf{x} \quad (3.5)$$

Eg. 1 点电荷的静电势

$$\phi(r) = \frac{q}{4\pi\varepsilon_0 r} \quad (3.6)$$

Eg. 2 均匀电场的静电势

设均匀电场 $\mathbf{E}_0$ 的零点为参考点, 则

$$\phi(\mathbf{x}) = -\mathbf{E}_0 \cdot \mathbf{x} = -E_0 x \cos \theta \quad (3.7)$$

Lem. 2

$$\int \frac{dz}{\sqrt{z^2 + R^2}} = \ln(z + \sqrt{z^2 + R^2}) + C \quad (3.8)$$

Eg. 3 无限长带电直线的静电势

设均匀带电无限长直线的电荷线密度为 λ , 则距离 R 远处的电势为

$$\phi(R) = -\frac{\lambda}{2\pi\epsilon_0} \ln \frac{R}{R_0} \quad (3.9)$$

Prop. 2 Poisson 方程

$$\nabla^2 \phi(\mathbf{x}) = -\frac{\rho(\mathbf{x})}{\epsilon} \quad (3.10)$$

对于存在自由电荷体分布的区域成立.

Prop. 3 Laplace 方程

$$\nabla^2 \phi(\mathbf{x}) = 0 \quad (3.11)$$

对于无自由电荷体分布的区域成立.

3.1.2 边值关系

Thm. 1 介质分界面的边值关系

$$\begin{cases} \phi_2 = \phi_1 \\ \epsilon_2 \frac{\partial \phi_2}{\partial n} - \epsilon_1 \frac{\partial \phi_1}{\partial n} = -\sigma \end{cases} \quad (3.12)$$

Pf $\phi_2 - \phi_1 = -\int_{p_1}^{p_2} \mathbf{E} \cdot d\mathbf{x} \rightarrow 0$, 因此介质分界面处电势连续, 有 $\phi_2 = \phi_1$. 设介质分界面存在自由电荷分布 σ , 则 $D_{2n} - D_{1n} = \epsilon_2 E_{2n} - \epsilon_1 E_{1n} = -\left(\epsilon_2 \frac{\partial \phi_2}{\partial n} - \epsilon_1 \frac{\partial \phi_1}{\partial n}\right) = \sigma$.

Thm. 2 绝缘介质分界面的边值关系

$$\begin{cases} \phi_2 = \phi_1 \\ \epsilon_2 \frac{\partial \phi_2}{\partial n} = \epsilon_1 \frac{\partial \phi_1}{\partial n} \end{cases} \quad (3.13)$$

Pf 绝缘介质分界面一般无自由电荷分布, 即 $\epsilon_2 E_{2n} = \epsilon_1 E_{1n}$, 因此 $\epsilon_2 \frac{\partial \phi_2}{\partial n} = \epsilon_1 \frac{\partial \phi_1}{\partial n}$.

Thm. 3 导体表面边值关系

设导体表面自由电荷分布为 σ , 外侧介质的介电常量为 ε , 则边值关系为

$$\begin{cases} \phi|_{\text{surf}} = \text{Const} \\ \frac{\partial \phi}{\partial n} = -\frac{\sigma}{\varepsilon} \end{cases} \quad (3.14)$$

Pf 由 $\mathbf{n} \times \mathbf{E} = 0$ 与 $\mathbf{n} \cdot \mathbf{D} = \sigma$ 易得.

Prop. 4 导体表面自由电荷量

若导体表面周围介质均匀, 整个导体表面总自由电荷量为

$$Q_f = \oint_{\text{surf}} \sigma d^2x = -\varepsilon \oint_{\text{surf}} \frac{\partial \phi}{\partial n} d^2x \quad (3.15)$$

Eg. 4 带电导体球壳的边界条件

设导体球壳带电 Q , 内外半径分别为 R_1, R_2 , 壳内外电势分别为 ϕ_1, ϕ_2 , 则

$$-\oint_{r=R_2} \frac{\partial \phi_2}{\partial r} d^2x + \oint_{r=R_1} \frac{\partial \phi_1}{\partial r} d^2x = \frac{Q}{\varepsilon_0} \quad (3.16)$$

Cor. 2 导体的静电条件

1. 内部无自由电荷, 电荷只分布于表面;
2. 内部电场为零;
3. 整个导体表面为等势面, 整个导体为等势体.

3.1.3 静电场的能量**Prop. 5 静电体系总能量**

$$W_\infty = \frac{1}{2} \int_\infty \rho(\mathbf{x}) \phi(\mathbf{x}) d^3x \quad (3.17)$$

Rmk 局部电场能量为 $W_\Omega = \frac{1}{2} \int_\Omega \mathbf{E} \cdot \mathbf{D} d^3x$, 能量密度 $w_e = \frac{\mathbf{E} \cdot \mathbf{D}}{2} \neq \frac{\rho \phi}{2}$.

Pf $W_\Omega = \frac{1}{2} \int_\Omega \mathbf{E} \cdot \mathbf{D} d^3x = -\frac{1}{2} \int_\Omega \nabla \phi \cdot \mathbf{D} d^3x = \frac{1}{2} \int_\Omega \rho \phi d^3x - \frac{1}{2} \int_{\partial\Omega} \phi \mathbf{D} \cdot d^2\mathbf{x}$, 当 $\Omega \rightarrow \infty, \phi D \sim r^{-3}, \partial\Omega \sim r^2, \Rightarrow W_\infty = \frac{1}{2} \int_\infty \rho \phi d^3x$.

Cor. 3 导体总能量

由于导体表面等势, 且电荷只分布于表面, 有

$$W_{\text{cond}} = \frac{1}{2} Q \phi \quad (3.18)$$

Lem. 3 Green 互易定理

$$\int \rho_1 \phi_2 \, d^3x = \int \rho_2 \phi_1 \, d^3x \quad (3.19)$$

Prop. 6 电荷体系间的相互作用能

$$W_{\text{int}} = \int_{\Omega'} \rho \phi_{\text{ext}} \, d^3x' \quad (3.20)$$

Pf 设 ρ, ρ_{ext} 产生的静电势分别为 ϕ, ϕ_{ext} . $W_{\text{total}} = \frac{1}{2} \int_{\Omega' + \Omega'_{\text{ext}}} (\rho + \rho_{\text{ext}})(\phi + \phi_{\text{ext}}) \, d^3x'$, 展开得 $W_{\text{total}} = \underbrace{\frac{1}{2} \int_{\Omega'} \rho \phi \, d^3x' + \frac{1}{2} \int_{\Omega'_{\text{ext}}} \rho_{\text{ext}} \phi_{\text{ext}} \, d^3x'}_{=W_{\text{self}}} + \underbrace{\frac{1}{2} \int_{\Omega'} \rho \phi_{\text{ext}} \, d^3x' + \frac{1}{2} \int_{\Omega'_{\text{ext}}} \rho_{\text{ext}} \phi \, d^3x'}_{=W_{\text{int}}}$, 结合 Green 互易定理得证.

3.1.4 Earnshaw 定理**Thm. 4 Earnshaw 定理**

仅靠静电力, 无法将粒子维持在稳定的力学平衡状态. 等价地, 静电势在有限无自由电荷分布的区域内, 不能达到极值.

3.2 静电场唯一性定理 ***Thm. 5 静电场唯一性定理**

已知区域 Ω 内的介质介电常量分布 $\varepsilon(\mathbf{x})$, 若给定

- Ω 内的自由电荷分布 $\rho(\mathbf{x})$;
- 边界 $\partial\Omega$ 上的
 - 电势 $\phi|_{\partial\Omega}$,
 - 电势的法向导数 $\left. \frac{\partial\phi}{\partial n} \right|_{\partial\Omega}$,

则 Ω 内的静电场唯一确定, 或 Ω 内 Poisson 方程的解在相差一个常量的意义下唯一.

3.3 镜像法

Meth. 1 镜像法

1. 使用条件

- (a) 求解区域的边界具有一定的对称性.
- (b) 求解区域内的自由电荷少量, 其电势 ϕ 可直接给出解析解.
- (c) 边界上感应面电荷或极化面电荷, 对求解区域的电势产生的效果, 可用镜像电荷代替.

2. 在求解区域外引入镜像电荷, 则求解区域的总电势为 $\phi_{\text{total}} = \phi + \phi'$.

3. 调整镜像电荷的位置与电荷量, 使总电势满足边界条件, 则根据静电场唯一性定理, ϕ_{total} 即所求 Poisson 方程的解;

4. 可进一步利用镜像电荷分析求解区域的 $\mathbf{E}$ 与 $\mathbf{F}$, 边界上的 σ .

Eg. 5 无限大接地导体平面

点电荷 q 距离无限大接地导体平面为 d , 设其位置为 $(0, 0, d)$, 则镜像电荷 $q' = -q$, 位置为 $(0, 0, -d)$. 求解域的电势为

$$\phi(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}} \right] \quad (3.21)$$

感应电荷面密度为

$$\sigma(x, y) = -\epsilon_0 \left. \frac{\partial \phi}{\partial z} \right|_{z=0} = -\frac{qd}{2\pi(x^2 + y^2 + d^2)^{3/2}} \quad (3.22)$$

总感应电荷

$$Q = \int_{-\infty}^{\infty} \sigma(x, y) dx dy = \int_{r=0}^{\infty} \int_{\theta=0}^{2\pi} \sigma(r, \theta) dr d\theta = -q = q' \quad (3.23)$$

点电荷受力为

$$\mathbf{F}(z) = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2z)^2} \mathbf{e}_z \quad (3.24)$$

Eg. 6 接地导体球

点电荷 q 距离半径为 R 的接地导体球球心为 $d(> R)$, 则镜像电荷 $q' = -\frac{R}{d}q$, 距离球心为

$d' = \frac{R^2}{d}$. 求解域电势为

$$\phi(r, \theta) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{r^2 + d^2 - 2rd \cos \theta}} - \frac{\frac{R}{d}q}{\sqrt{r^2 + \left(\frac{R^2}{d}\right)^2 - 2r\frac{R^2}{d} \cos \theta}} \right] \quad (3.25)$$

若导体球不接地, 而带电荷 Q_0 , 则感应电荷可等效为两个镜像电荷的叠加:

1. $q' = -\frac{R}{d}q$, 距离球心为 $d' = \frac{R^2}{d}$;
2. $q'' = Q_0 - q'$, 位于球心.

求解域电势为

$$\phi(r, \theta) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{r^2 + d^2 - 2rd \cos \theta}} - \frac{\frac{R}{d}q}{\sqrt{r^2 + \left(\frac{R^2}{d}\right)^2 - 2r\frac{R^2}{d} \cos \theta}} + \frac{Q_0 + \frac{R}{d}Q}{r} \right] \quad (3.26)$$

Eg. 7 组合接地边界

组合边界通常由平面, 球面, 柱面 (可视为二维球面) 的部分组合而成, 需进行多次镜像.

1. 球面或柱面 + 球面或柱面: 通常不用镜像法求解.
2. 两相交平面: 夹角为 π/n 且电荷位于角平分面上时可解.
3. 平面 + 球面或柱面: 球心或圆柱轴线位于平面上时可解. 先对球面或柱面做镜像 q' , 再将 q 与第一次的镜像电荷 q' 对平面做镜像, 得到 $-q$ 和 $-q'$.

3.4 Green 函数法

3.4.1 解的 Green 函数表示

Def. 1 Green 函数

Green 函数为单位点源影响函数

$$\nabla^2 G(\mathbf{x}, \mathbf{x}') = -\frac{\delta(\mathbf{x} - \mathbf{x}')}{\epsilon} \quad (3.27)$$

满足第一类或第二类边界条件

$$G_D(\mathbf{x}, \mathbf{x}')|_S = 0 \quad (3.28)$$

$$\left. \frac{\partial G_N(\mathbf{x}, \mathbf{x}')}{\partial n} \right|_S = -\frac{1}{\varepsilon S} \quad (3.29)$$

Lem. 4 第一 Green 公式

$$\int_{\Omega} \phi \nabla^2 \psi + \nabla \phi \cdot \nabla \psi \, d^3x = \oint_{\partial\Omega} \phi \nabla \psi \cdot d^2\mathbf{x} \quad (3.30)$$

Pf $\nabla \cdot (\phi \nabla \psi) = \nabla \phi \cdot \nabla \psi + \phi \nabla^2 \psi.$

Lem. 5 第二 Green 公式

$$\int_{\Omega} \phi \nabla^2 \psi - \psi \nabla^2 \phi \, d^3x = \oint_{\partial\Omega} (\phi \nabla \psi - \psi \nabla \phi) \cdot d^2\mathbf{x} \quad (3.31)$$

Pf $\nabla \cdot (\phi \nabla \psi - \psi \nabla \phi) = (\nabla \phi \cdot \nabla \psi + \phi \nabla^2 \psi) - (\nabla \psi \cdot \nabla \phi + \psi \nabla^2 \phi) = \phi \nabla^2 \psi - \psi \nabla^2 \phi.$

Prop. 7 解的表示

对于各向同性的线性均匀介质, Poisson 方程的解为

$$\phi(\mathbf{x}) = \int_{\Omega'} \rho(\mathbf{x}') G(\mathbf{x}', \mathbf{x}) \, d^3x' + \varepsilon \oint_{\partial\Omega'} G \frac{\partial \phi}{\partial n'} - \phi \frac{\partial G}{\partial n'} \, d^2x' \quad (3.32)$$

Rmk 选取 $G(\mathbf{x}', \mathbf{x})$ 的边界条件, 可使得 $\phi(\mathbf{x})$ 只与 $\phi(\mathbf{x}')$, $\frac{\partial \phi(\mathbf{x}')}{\partial n'}$ 其一相关.

Pf 第二 Green 公式中令 $\psi = G$.

$$\begin{aligned} \Rightarrow \int_{\Omega} \phi \nabla^2 G - G \nabla^2 \phi \, d^3x &= \int_{\Omega} \frac{G(\mathbf{x}, \mathbf{x}') \rho(\mathbf{x})}{\varepsilon} - \frac{\phi(\mathbf{x}) \delta(\mathbf{r})}{\varepsilon} \, d^3x = \frac{1}{\varepsilon} \int_{\Omega} \rho(\mathbf{x}) G(\mathbf{x}, \mathbf{x}') \, d^3x - \frac{\phi(\mathbf{x}')}{\varepsilon} \\ &\Rightarrow \phi(\mathbf{x}') = \int_{\Omega} \rho(\mathbf{x}) G(\mathbf{x}, \mathbf{x}') \, d^3x + \varepsilon \oint_{\partial\Omega} G \frac{\partial \phi}{\partial n} - \phi \frac{\partial G}{\partial n} \, d^2x \end{aligned}$$

做场源互换 $\mathbf{x} \leftrightarrow \mathbf{x}'$, 由 Green 函数对场点与源点的对称性, 得证.

3.4.2 Dirichlet 边值问题

3.4.3 Neumann 边界问题

3.4.4 几类特殊边值问题的 Green 函数

3.5 Laplace 方程的分离变量法

Meth. 2 Poisson 方程问题的简化

若已知求解域内的电荷产生的电势为 ϕ_f , 则可令 Poisson 方程的解为

$$\phi = \phi_f + \phi' \quad (3.33)$$

其中 ϕ' 满足 Laplace 方程. 可用分离变量法求解 ϕ' , 再利用解的叠加原理得到 ϕ .

3.5.1 球域问题

Lem. 6 球坐标下的 Laplace 方程

$$\nabla^2 \phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \phi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \phi}{\partial \varphi^2} = 0 \quad (3.34)$$

Thm. 6 球域 Laplace 方程的通解

设电势为 $\phi = \phi(r, \theta, \varphi) = R(r)\Theta(\theta)\Phi(\varphi)$, 可得通解

$$\phi = \sum_{n=0}^{\infty} \sum_{m=0}^n \left[\left(A_n^m r^n + \frac{B_n^m}{r^{n+1}} \right) P_n^m(\cos \theta) \cos m\varphi + \left(C_n^m r^n + \frac{D_n^m}{r^{n+1}} \right) P_n^m(\cos \theta) \sin m\varphi \right] \quad (3.35)$$

Thm. 7 轴对称通解

若 $\frac{\partial \phi}{\partial \varphi} = 0$, 取 $m = 0$ 可得轴对称通解

$$\phi = \sum_{n=0}^{\infty} \left(A_n r^n + \frac{B_n}{r^{n+1}} \right) P_n(\cos \theta) \quad (3.36)$$

Thm. 8 球对称通解

若 $\frac{\partial \phi}{\partial \theta} = \frac{\partial \phi}{\partial \varphi} = 0$, 取 $n = m = 0$ 可得球对称通解

$$\phi = A + \frac{B}{r} \quad (3.37)$$

3.5.2 圆柱域问题 *

3.5.3 保角变换 *

3.5.4 圆域问题

Lem. 7 极坐标下的 Laplace 方程

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0 \quad (3.38)$$

Thm. 9 圆域 Laplace 方程的通解

设电势为 $\phi(r, \theta) = R(r)\Theta(\theta)$, 可得通解

$$\phi = A_0 + B_0 \ln r + \sum_{n=1}^{\infty} (A_n r^n + B_n r^{-n}) (C_n \cos n\theta + D_n \sin n\theta) \quad (3.39)$$

3.6 电多极矩 *

多级展开条件: 源相对集中于一个小区域, 且线度远远小于源点到场点的距离.

3.6.1 静电势的多极展开

$$\phi(\mathbf{x}) \quad (3.40)$$

3.6.2 电偶极矩

Prop. 8 电偶极子的电势

$$\phi_p = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \mathbf{r}}{r^3} \quad (3.41)$$

Prop. 9 电偶极子的电场

$$\mathbf{E}_p = \frac{1}{4\pi\epsilon_0} \frac{3(\mathbf{p} \cdot \mathbf{e}_r)\mathbf{e}_r - \mathbf{p}}{r^3} \quad (3.42)$$

Prop. 10 电偶极子的电荷密度

$$\rho_p = -\mathbf{p} \cdot \nabla \delta(\mathbf{r}) \quad (3.43)$$

3.6.3 电势极矩

$$D_{\alpha\beta} = \int (3x'_\alpha x'_\beta - r'^2 \delta_{\alpha\beta}) \rho(\mathbf{x}') d^3x' \quad (3.44)$$

3.6.4 静电能量的多极展开

3.6.5 球域中的电多极矩

Lem. 8 球谐函数的加法公式

$$P_l(\cos \gamma) = \sum_{m=-l}^{+l} \frac{(l-m)!}{(l+m)!} P_l^m(\cos \theta) P_l^m(\cos \theta') e^{im(\varphi - \varphi')} \quad (3.45)$$

其中 $\cos \gamma = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\varphi - \varphi')$.

$$A_{lm} = \frac{4\pi}{2l+1} \int_{V'} \rho(\mathbf{x}') x'^l Y_{lm}^*(\theta', \varphi') \quad (3.46)$$

$$B_{lm} = \frac{4\pi}{2l+1} \int_{V'} \rho(\mathbf{x}') \frac{1}{x'^{l+1}} Y_{lm}^*(\theta', \varphi') \quad (3.47)$$

4 静磁学

Lem. 1 静磁场基本方程

$$\begin{cases} \nabla \times \mathbf{H} = \mathbf{J} \\ \nabla \cdot \mathbf{B} = 0 \\ \mathbf{B} = \mu \mathbf{H} \\ \begin{cases} \mathbf{n} \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0 \\ \mathbf{n} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{K} \end{cases} \end{cases} \quad (4.1)$$

4.1 磁矢势

4.1.1 磁矢势的微分方程

Prop. 1 磁感应强度的磁矢势表示

$$\nabla \times \mathbf{A} = \mathbf{B} \quad (4.2)$$

Thm. 1 磁矢势的规范不变性

磁矢势满足规范不变性

$$\mathbf{B}' = \nabla \times \mathbf{A}' = \nabla \times (\mathbf{A} + \nabla \chi) = \nabla \times \mathbf{A} + \underbrace{\nabla \times (\nabla \chi)}_{=0} = \mathbf{B} \quad (4.3)$$

一般静磁场采用 Coulomb 规范

$$\nabla \cdot \mathbf{A}(\mathbf{x}) = 0 \quad (4.4)$$

Thm. 2 磁矢势的 Poisson 方程

$$\nabla^2 \mathbf{A} = -\mu \mathbf{J} \quad (4.5)$$

$$\text{Pf} \quad \nabla \times \mathbf{B} = \mu \nabla \times \mathbf{H} = \mu \mathbf{J} = \nabla \times (\nabla \times \mathbf{A}) = \underbrace{\nabla (\nabla \cdot \mathbf{A})}_{=0} - \nabla^2 \mathbf{A} = -\mu \mathbf{J}.$$

Eg. 1 无限长带电直导线

无限长带电直导线沿 z 轴通有电流 I , 设距离导线 r_0 处为令矢势面, 则 r 处的矢势为

$$\mathbf{A}(r) = -\frac{\mu_0 I}{2\pi} \ln \frac{r}{r_0} \mathbf{e}_z \quad (4.6)$$

Eg. 2 均匀磁场

设 $\mathbf{B} = B_0 \mathbf{e}_z$, 则矢势为 $\forall \alpha \in \mathbb{R}$

$$\mathbf{A}(x, y) = [-\alpha y \mathbf{e}_x + (1 - \alpha)x \mathbf{e}_y] B_0 \quad (4.7)$$

4.1.2 边值关系

Thm. 3 磁矢势的边值关系

$$\begin{cases} A_{2t} = A_{1t} \xrightarrow{\text{Coulomb 规范}} \mathbf{A}_2 = \mathbf{A}_1 \\ \mathbf{n} \times \left(\nabla \times \frac{\mathbf{A}_2}{\mu_2} - \nabla \times \frac{\mathbf{A}_1}{\mu_1} \right) = \mathbf{K} \end{cases} \quad (4.8)$$

$$\begin{aligned} \text{Pf } \oint_{\partial \Sigma} \mathbf{A} \cdot d\mathbf{x} &= \int_{\Sigma} \mathbf{B} \cdot d^2\mathbf{x} \rightarrow 0, \Rightarrow A_{2t} = A_{1t}, \nabla \cdot \mathbf{A} = 0, \Rightarrow A_{2n} = A_{1n}, \Rightarrow \mathbf{A}_2 = \mathbf{A}_1. \\ \mathbf{K} &= \mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{n} \times \left(\nabla \times \frac{\mathbf{A}_2}{\mu_2} - \nabla \times \frac{\mathbf{A}_1}{\mu_1} \right). \end{aligned}$$

4.1.3 静磁场的能量

Prop. 2 磁场的总能量

设 Ω 为电流分布区域, 则

$$W = \frac{1}{2} \int_{\Omega} \mathbf{A} \cdot \mathbf{J} d^3x \quad (4.9)$$

$$\text{Pf } \mathbf{B} \cdot \mathbf{H} = \nabla \cdot (\mathbf{A} \times \mathbf{H}) + \mathbf{A} \cdot \mathbf{J}, \Rightarrow W_{\infty} = \frac{1}{2} \int_{\infty} \mathbf{B} \cdot \mathbf{H} d^3x = \frac{1}{2} \int_{\Omega} \mathbf{A} \cdot \mathbf{J} d^3x.$$

Eg. 3 无限长载流线圈总磁能

$$W = \frac{1}{2} I \Phi \quad (4.10)$$

Prop. 3 磁相互作用能

$$W_{\text{int}} = \int_{\Omega'} \mathbf{A}_{\text{ext}} \cdot \mathbf{J} d^3x' - W_{\varepsilon} \quad (4.11)$$

其中 W_{ε} 为外电源为了维持电流体系恒定, 而额外做的功.

4.1.4 Aharonov-Bohm 效应

Thm. 4 A-B 效应本质

粒子运动轨迹所围的电磁通量改变了粒子波函数的相位差.

螺线管未通电时, 两束电子波函数相位差为

$$\varphi_0 \approx k_0 d \sin \theta \quad (4.12)$$

通电后相位差为

$$\varphi_1 = \varphi_0 - \frac{e}{\hbar} \oint_C \mathbf{A} \cdot d\mathbf{x} \quad (4.13)$$

相位差的改变量为

$$\Delta\varphi = \varphi_1 - \varphi_0 = -\frac{e}{\hbar} \Phi \quad (4.14)$$

n 级干涉亮纹移动

$$\Delta x = f \Delta(\tan \theta) \approx \frac{e\Phi f}{mvd} \quad (4.15)$$

4.2 静磁场唯一性定理 *

Thm. 5 静磁场唯一性定理

4.3 磁标势

Prop. 4 引入磁标势的条件

若区域 Ω 内不存在自由电流或传导电流体分布, 且其边界面 $\partial\Omega$ 将 Ω 分隔为一个单连通区域, 即任意回路都不被自由电流所链环

$$\begin{cases} \nabla \times \mathbf{H} = 0 \\ \oint_{\forall L \subset \Omega} \mathbf{H} \cdot d\mathbf{x} = 0 \end{cases} \quad (4.16)$$

则可在 Ω 中定义标量势 ϕ_m 使得

$$\mathbf{H} = -\nabla \phi_m \quad (4.17)$$

对于非铁磁的线性介质

$$\mathbf{B} = -\mu \nabla \phi_m \quad (4.18)$$

Thm. 6 磁标势的 Poisson 方程

定义等效磁荷 $\rho_m = -\mu_0 \nabla \cdot \mathbf{M}$, 则有

$$\nabla^2 \phi_m = -\frac{\rho_m}{\mu_0} \quad (4.19)$$

$$\text{Pf } \nabla \cdot \mathbf{B} = \mu_0 \nabla \cdot (\mathbf{H} + \mathbf{M}) = 0, \Rightarrow \nabla \cdot \mathbf{H} = -\nabla^2 \phi_m = -\nabla \cdot \mathbf{M} = \frac{\rho_m}{\mu_0}$$

Thm. 7 磁标势的边值关系

由 $H_{1t} = H_{2t}$, $B_{1n} = B_{2n}$ 可得

$$\begin{cases} \phi_1 = \phi_2 \\ \frac{\partial \phi_1}{\partial n} - \frac{\partial \phi_2}{\partial n} = \mathbf{n} \cdot (\mathbf{M}_1 - \mathbf{M}_2) \end{cases} \quad (4.20)$$

对于非铁磁各向同性介质, 由 $\mathbf{B} = \mu \mathbf{H}$ 可得

$$\begin{cases} \phi_1 = \phi_2 \\ \mu_1 \frac{\partial \phi_1}{\partial n} = \mu_2 \frac{\partial \phi_2}{\partial n} \end{cases} \quad (4.21)$$

静电场与静磁场的对比

$$\begin{aligned} \mathbf{E} &= -\nabla \phi & \mathbf{H} &= -\nabla \phi_m \\ \nabla \cdot \mathbf{E} &= \frac{\rho_f + \rho_P}{\varepsilon_0} & \nabla \cdot \mathbf{H} &= \frac{\rho_m}{\mu_0} \\ \rho_P &= -\nabla \cdot \mathbf{P} & \rho_m &= -\mu_0 \nabla \cdot \mathbf{M} \\ \nabla^2 \phi &= -\frac{\rho + \rho_P}{\varepsilon_0} & \nabla^2 \phi_m &= -\frac{\rho_m}{\mu_0} \\ \begin{cases} \phi_2 = \phi_1 \\ \varepsilon_2 \frac{\partial \phi_2}{\partial n} = \varepsilon_1 \frac{\partial \phi_1}{\partial n} \end{cases} & & \begin{cases} \phi_2 = \phi_1 \\ \mu_2 \frac{\partial \phi_2}{\partial n} = \mu_1 \frac{\partial \phi_1}{\partial n} \end{cases} \end{aligned}$$

4.4 磁多极矩 *

Thm. 8 磁矢势的多极展开

$$\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}')}{r} d^3x' = \mathbf{A}^{(0)}(\mathbf{x}) + \mathbf{A}^{(1)}(\mathbf{x}) + \mathbf{A}^{(2)}(\mathbf{x}) + \dots \quad (4.22)$$

$$\mathbf{A}^{(0)}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}')}{x} d^3x' \quad (4.23)$$

$$\mathbf{A}^{(1)}(\mathbf{x}) = -\frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{x}') (\mathbf{x}' \cdot \nabla) \frac{1}{x} d^3x' \quad (4.24)$$

$$\mathbf{A}^{(2)}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{x}') \frac{1}{2} \mathbf{x}' \mathbf{x}' : \nabla \nabla \frac{1}{x} d^3x' \quad (4.25)$$

5 电磁波

5.1 电磁波基本方程

Lem. 1 电磁波波动方程

在各向同性的线性均匀非色散介质中

$$\begin{cases} \left(\nabla^2 - \mu\varepsilon \frac{\partial^2}{\partial t^2} \right) \mathbf{E}(t, \mathbf{x}) = 0 \\ \left(\nabla^2 - \mu\varepsilon \frac{\partial^2}{\partial t^2} \right) \mathbf{B}(t, \mathbf{x}) = 0 \end{cases} \quad (5.1)$$

Rmk 可得电磁波的传播速度为 $v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{\sqrt{\mu_r \varepsilon_r}} = \frac{c}{n}$.

5.1.1 时谐电磁波

Def. 1 时谐电磁波

时谐电磁波的场量为

$$\mathbf{E}(t, \mathbf{x}) = \mathbf{E}(\mathbf{x})e^{-i\omega t} \quad \mathbf{B}(t, \mathbf{x}) = \mathbf{B}(\mathbf{x})e^{-i\omega t} \quad (5.2)$$

Thm. 1 Helmholtz 方程

时谐电磁波的波动方程可由满足 $\nabla \cdot \mathbf{E} = 0$, $\nabla \cdot \mathbf{B} = 0$ 的 Helmholtz 方程描述

$$\nabla^2 \mathbf{E}(\mathbf{x}) + k^2 \mathbf{E}(\mathbf{x}) = 0 \quad \nabla^2 \mathbf{B}(\mathbf{x}) + k^2 \mathbf{B}(\mathbf{x}) = 0 \quad (5.3)$$

其中 $k = \omega\sqrt{\mu\varepsilon} = \frac{\omega}{v}$ 为波数.

$$\text{Pf} \quad \frac{\partial}{\partial t} \mathbf{E}(t, \mathbf{x}) = -i\omega \mathbf{E}(\mathbf{x}), \quad \frac{\partial^2}{\partial t^2} \mathbf{E}(t, \mathbf{x}) = -\omega^2 \mathbf{E}(\mathbf{x}).$$

Prop. 1 电场与磁场的关系

$$\nabla \times \mathbf{E}(\mathbf{x}) = i\omega \mathbf{B}(\mathbf{x}) \quad \mathbf{E}(\mathbf{x}) = i\frac{v}{k} \nabla \times \mathbf{B}(\mathbf{x}) \quad (5.4)$$

$$\text{Pf} \quad \nabla \times \mathbf{E}(t, \mathbf{x}) = e^{-i\omega t} \nabla \times \mathbf{E}(\mathbf{x}) = -\frac{\partial}{\partial t} \mathbf{B}(t, \mathbf{x}) = i\omega \mathbf{B}(\mathbf{x})e^{-i\omega t}.$$

$$\nabla \times \mathbf{B}(t, \mathbf{x}) = e^{-i\omega t} \nabla \times \mathbf{B}(\mathbf{x}) = \mu\varepsilon \frac{\partial}{\partial t} \mathbf{E}(t, \mathbf{x}) = -i\omega\mu\varepsilon \mathbf{E}(\mathbf{x})e^{-i\omega t}, \text{ where } \omega\mu\varepsilon = \frac{\omega}{v^2} = \frac{k}{v}.$$

5.1.2 平面单色波

Def. 2 平面单色波

平面单色波的场量为

$$\mathbf{E}(t, \mathbf{x}) = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \quad \mathbf{B}(t, \mathbf{x}) = \mathbf{B}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \quad (5.5)$$

平面波的相位为

$$\varphi = \mathbf{k} \cdot \mathbf{x} - \omega t + \varphi_0 \quad (5.6)$$

其中初相位因子 $e^{i\varphi_0}$ 一般并入复振幅 $\mathbf{E}_0, \mathbf{B}_0$ 中.

Def. 3 相速度

同一时刻 φ 相同的点构成的称为等相面. 电磁波等相面的运动速度, 称为相速度.

$$\mathbf{v}_p = v \mathbf{e}_k = \frac{\omega}{k} \mathbf{e}_k \quad (5.7)$$

Def. 4 群速度

色散关系描述了频率与波矢的关系

$$k = \pm \omega \sqrt{\mu \varepsilon} \quad (5.8)$$

定义相速度为

$$v_g = \frac{d\omega}{dk} \quad (5.9)$$

Prop. 2 平面单色波的性质

- 为 TEM 横电磁模波, 即 $\mathbf{E} \perp \mathbf{k}, \mathbf{B} \perp \mathbf{k}, \mathbf{E} \perp \mathbf{B}$.

- $\mathbf{E}(\mathbf{x})$ 与 $\mathbf{B}(\mathbf{x})$ 满足

$$\mathbf{E}(\mathbf{x}) = v \mathbf{B}(\mathbf{x}) \times \mathbf{e}_k \quad (5.10)$$

- $\mathbf{E}, \mathbf{B}$ 同相位, 振幅之比为

$$\frac{E_0}{B_0} = v \quad (5.11)$$

其中 $E_0 = \|\mathbf{E}_0\|, B_0 = \|\mathbf{B}_0\|$.

- 群速度等于相速度, 即

$$v_g = \frac{d}{dk} \frac{k}{\sqrt{\mu \varepsilon}} = \frac{1}{\sqrt{\mu \varepsilon}} = v_p \quad (5.12)$$

Pf $\nabla \cdot \mathbf{E}(t, \mathbf{x}) = \mathbf{E}_0 \cdot \nabla e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = i \mathbf{k} \cdot \mathbf{E} = 0$, 同理 $\mathbf{k} \cdot \mathbf{B} = 0$.

$$\mathbf{E}(\mathbf{x}) = i \frac{v}{k} \nabla \times (\mathbf{B}_0 e^{i\mathbf{k} \cdot \mathbf{x}}) = i \frac{v}{k} \underbrace{\nabla e^{i\mathbf{k} \cdot \mathbf{x}}}_{= i\mathbf{k} e^{i\mathbf{k} \cdot \mathbf{x}}} \times \mathbf{B}_0 = v \mathbf{B} \times \mathbf{e}_k.$$

Def. 5 电磁波阻抗

$$Z = \sqrt{\frac{\mu}{\varepsilon}} = Z_0 \sqrt{\frac{\mu_r}{\varepsilon_r}} \quad (5.13)$$

其中 $Z_0 = \sqrt{\mu_0/\varepsilon_0} \approx 377 \, \Omega$ 为真空中波阻抗.

Rmk $Z\mathbf{H} = \mathbf{e}_k \times \mathbf{E}$, $E_0/H_0 = Z$.

5.1.3 能量与能流**Prop. 3 平面单色波的能量密度**

平面电磁波的电场能量与磁场能量相等.

$$\begin{aligned} w(t, \mathbf{x}) &= \frac{1}{2} \varepsilon E^2 + \frac{1}{2\mu} B^2 \\ &= \varepsilon E^2 = \varepsilon E_0^2 \cos^2(\mathbf{k} \cdot \mathbf{x} - \omega t) \\ &= \frac{1}{\mu} B^2 = \frac{1}{\mu} B_0^2 \cos^2(\mathbf{k} \cdot \mathbf{x} - \omega t) \end{aligned} \quad (5.14)$$

$$\langle w \rangle = \int_0^T w(t, \mathbf{x}) dt = \frac{1}{2} \varepsilon E_0^2 = \frac{1}{2\mu} B_0^2 \quad (5.15)$$

Prop. 4 平面单色波的能流密度

$$\mathbf{S}(t, \mathbf{x}) = \mathbf{E} \times \mathbf{H} = \frac{1}{Z} E^2 \mathbf{e}_k = v_p w \mathbf{e}_k \quad (5.16)$$

$$\langle \mathbf{S} \rangle = \frac{1}{2Z} E_0^2 \mathbf{e}_k = v_p \langle w \rangle \mathbf{e}_k \quad (5.17)$$

Def. 6 能量传播速度

$$\mathbf{v}_e = \frac{\langle \mathbf{S} \rangle}{\langle w \rangle} \quad (5.18)$$

Rmk 对于平面单色波 $\mathbf{v}_e = \mathbf{v}_p = \mathbf{v}_g$.

5.2 偏振**Def. 7 偏振方向**

由于 $\mathbf{E} \perp \mathbf{k}$, 取 $\mathbf{E}$ 的振荡方向为电磁波的偏振方向.

Thm. 2 电矢量的分解

选取标准正交基 $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, 令 $\mathbf{e}_k = \mathbf{e}_3$, 设相位 $\varphi = \mathbf{k} \cdot \mathbf{x} - \omega t$, 可将复电矢量分解为

$$\tilde{\mathbf{E}}(t, \mathbf{x}) = (E_{10}e^{i\delta_1}\mathbf{e}_1 + E_{20}e^{i\delta_2}\mathbf{e}_2)e^{i\varphi} \quad (5.19)$$

5.2.1 椭圆偏振**Prop. 5 椭圆偏振方程**

设 $\mathbf{E} = E_1\mathbf{e}_1 + E_2\mathbf{e}_2$, 令 $\delta = \delta_2 - \delta_1$, 有电矢量末端点在 $\mathbf{e}_1$ - $\mathbf{e}_2$ 平面形成的椭圆轨迹方程

$$\left(\frac{E_1}{E_{10}}\right)^2 + \left(\frac{E_2}{E_{20}}\right)^2 - 2\frac{E_1E_2}{E_{10}E_{20}}\cos\delta = \sin^2\delta \quad (5.20)$$

Pf

$$\mathbf{E} = \text{Re}\{\tilde{\mathbf{E}}\} = E_{10}\cos(\delta_1 + \varphi)\mathbf{e}_1 + E_{20}\cos(\delta_2 + \varphi)\mathbf{e}_2 = E_1\mathbf{e}_1 + E_2\mathbf{e}_2$$

$$\Rightarrow \begin{cases} \frac{E_1}{E_{10}} = \cos(\delta_1 + \varphi) \equiv X & \Rightarrow \varphi = \arccos X - \delta_1 \\ \frac{E_2}{E_{20}} = \cos(\delta_2 + \varphi) \equiv Y & \Rightarrow \varphi = \arccos Y - \delta_2 \end{cases}$$

$$\Rightarrow \arccos Y - \arccos X = \delta \xrightarrow{\sin(\arccos x) = \sqrt{1-x^2}} \begin{cases} \cos\delta = XY + \sqrt{1-X^2}\sqrt{1-Y^2} \\ \sin\delta = X\sqrt{1-Y^2} - Y\sqrt{1-X^2} \end{cases}$$

消去含根号的项: $\sin^2\delta + 2XY\cos\delta = X^2 + Y^2$ 整理可得一般椭圆标准方程

5.2.2 线偏振**Prop. 6 线偏振**

若 $\delta = m\pi$ ($m \in \mathbb{Z}$), 则椭圆偏振变为线偏振

$$\mathbf{E} = (E_{10}\mathbf{e}_1 \pm E_{20}\mathbf{e}_2)\cos(\mathbf{k} \cdot \mathbf{x} - \omega t + \delta_1) \quad (5.21)$$

m 为偶数时取 $+$, 奇数时取 $-$.

5.2.3 圆偏振

若 $\delta = \frac{\pi}{2} + m\pi$ ($m \in \mathbb{Z}$) 且 $E_{10} = E_{20} = \frac{E_0}{\sqrt{2}}$, 则椭圆偏振变为圆偏振

$$\begin{cases} \mathbf{E}_L = \frac{E_0}{\sqrt{2}} [\cos(\mathbf{k} \cdot \mathbf{x} - \omega t + \delta_1)\mathbf{e}_1 + \sin(\mathbf{k} \cdot \mathbf{x} - \omega t + \delta_1)\mathbf{e}_2] & \text{左旋圆偏振} \\ \mathbf{E}_R = \frac{E_0}{\sqrt{2}} [\cos(\mathbf{k} \cdot \mathbf{x} - \omega t + \delta_1)\mathbf{e}_1 - \sin(\mathbf{k} \cdot \mathbf{x} - \omega t + \delta_1)\mathbf{e}_2] & \text{右旋圆偏振} \end{cases} \quad (5.22)$$

Def. 8 圆偏振基矢

$$\mathbf{e}_{\pm} = \frac{1}{\sqrt{2}}(\mathbf{e}_1 \pm i\mathbf{e}_2) \quad (5.23)$$

$$\Rightarrow \begin{cases} \tilde{\mathbf{E}}_L = \frac{E_0}{\sqrt{2}}(\mathbf{e}_1 + i\mathbf{e}_2)e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = \tilde{E}_0 \mathbf{e}_+ \\ \tilde{\mathbf{E}}_R = \frac{E_0}{\sqrt{2}}(\mathbf{e}_1 - i\mathbf{e}_2)e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = \tilde{E}_0 \mathbf{e}_- \end{cases} \quad (5.24)$$

$\{\mathbf{e}_{\pm}\}$ 有如下性质

$$\begin{cases} \mathbf{e}_{\pm} \cdot \mathbf{e}_3 = 0 \\ \mathbf{e}_{\pm} \cdot \mathbf{e}_{\mp}^* = \mathbf{e}_{\mp} \cdot \mathbf{e}_{\pm}^* = 0 \quad \text{正交性} \\ \mathbf{e}_{\pm} \cdot \mathbf{e}_{\pm}^* = 1 \quad \text{归一性} \end{cases} \quad (5.25)$$

5.2.4 Stokes 参数与偏振度

Def. 9 Stokes 参数

设 $\mathbf{e}_1 \cdot \tilde{\mathbf{E}}_0 = a_1 e^{i\delta_1}$, $\mathbf{e}_2 \cdot \tilde{\mathbf{E}}_0 = a_2 e^{i\delta_2}$ ($a_1, a_2 \in \mathbb{R}^+$), 则 Stokes 参数为

$$s_0 = a_1^2 + a_2^2 \quad s_1 = a_1^2 - a_2^2 \quad s_2 = 2a_1 a_2 \cos(\delta_2 - \delta_1) \quad s_3 = 2a_1 a_2 \sin(\delta_2 - \delta_1) \quad (5.26)$$

Def. 10 偏振度

偏振度定义为 ($0 \leq P \leq 1$):

$$P = \frac{\sqrt{s_1^2 + s_2^2 + s_3^2}}{s_0} \quad (5.27)$$

Prop. 7

1. 对于完全偏振光 ($P = 1$), 满足

$$s_0^2 = s_1^2 + s_2^2 + s_3^2 \quad (5.28)$$

2. 对于部分偏振光, 有 $s_0^2 > s_1^2 + s_2^2 + s_3^2$, $0 < P < 1$.

3. 对于完全非偏振光 (自然光), 则 $s_1 = s_2 = s_3 = 0$, $P = 0$.

5.3 电磁波的衍射

5.3.1 Kirchhoff 公式

Thm. 3 Kirchhoff 公式

$$\psi(\mathbf{x}) = -\frac{1}{4\pi} \oint_{\Sigma} \frac{e^{ikr}}{r} \left[\nabla' \psi + \left(ik - \frac{1}{r} \right) \frac{\mathbf{r}}{r} \psi \right] \cdot d^2 \mathbf{x} \quad (5.29)$$

5.3.2 Fraunhofer 衍射

5.4 电磁波在介质界面上的反射与折射

5.4.1 角度关系

Thm. 4 反射定律与 Snell 折射定律

$$\theta = \theta' \quad (5.30)$$

$$n_1 \sin \theta = n_2 \sin \theta'' \quad (5.31)$$

5.4.2 Fresnel 公式

Thm. 5 Fresnel 公式

$$\left\{ \begin{array}{l} \left(\frac{E'}{E} \right)_p = \frac{\tan(\theta - \theta'')}{\tan(\theta + \theta'')} \quad \left(\frac{E'}{E} \right)_s = -\frac{\sin(\theta - \theta'')}{\sin(\theta + \theta'')} \\ \left(\frac{E''}{E} \right)_p = \frac{2 \cos \theta \sin \theta''}{\sin(\theta + \theta'') \cos(\theta - \theta'')} \quad \left(\frac{E''}{E} \right)_s = \frac{2 \cos \theta \sin \theta''}{\sin(\theta + \theta'')} \end{array} \right\} \quad (5.32)$$

$$\left\{ \begin{array}{l} \left(\frac{E'_0}{E_0} \right)_p = \frac{Z_1 \cos \theta - Z_2 \cos \theta''}{Z_1 \cos \theta + Z_2 \cos \theta''} \quad \left(\frac{E'_0}{E_0} \right)_s = \frac{Z_2 \cos \theta - Z_1 \cos \theta''}{Z_2 \cos \theta + Z_1 \cos \theta''} \\ \left(\frac{E''_0}{E_0} \right)_p = \frac{2Z_2 \cos \theta}{Z_1 \cos \theta + Z_2 \cos \theta''} \quad \left(\frac{E''_0}{E_0} \right)_s = \frac{2Z_2 \cos \theta}{Z_2 \cos \theta + Z_1 \cos \theta''} \end{array} \right\} \quad (5.33)$$

Cor. 1 垂直入射时电场振幅比

垂直入射时 p 偏振与 s 偏振无区别.

$$\left\{ \frac{E'_0}{E_0} = \frac{Z_2 - Z_1}{Z_2 + Z_1} \quad \frac{E''_0}{E_0} = \frac{2Z_2}{Z_2 + Z_1} \right\} \quad (5.34)$$

Prop. 8 阻抗匹配

5.4.3 Brewster 角

Thm. 6 Brewster 角

对于 p 偏振, 当入射角 $\theta = \theta_B$ 时, 反射系数为零 R_p . 其中 Brewster 角 θ_B 满足

$$\tan \theta_B = \sqrt{\frac{(Z_1/Z_2)^2 - 1}{1 - (n_1/n_2)^2}} = \sqrt{\frac{\varepsilon_2 \varepsilon_2 \mu_1 - \varepsilon_1 \mu_2}{\varepsilon_1 \varepsilon_2 \mu_2 - \varepsilon_1 \mu_1}} \quad (5.35)$$

Cor. 2 非磁性介质的 Brewster 角

$$\tan \theta_B = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}} = \frac{n_2}{n_1} \quad (5.36)$$

5.4.4 半波损失

5.4.5 全反射

Prop. 9 全反射

若 $n_1 > n_2$, 定义临界角 θ_c 为

$$\sin \theta_c = \frac{n_2}{n_1} \quad (5.37)$$

当 $\theta = \theta_c$ 时, $\theta'' = 90^\circ$. 当 $\sin \theta > \frac{n_2}{n_1}$ 时, 无法定义实数折射角, 称为全反射现象.

Prop. 10 消逝波

$\theta > \theta_c$ 时, 有 $k_1'' = k \sin \theta$, $k'' = k \sin \theta_c$, 因此

$$k_3'' = ik_1 \sqrt{\sin^2 \theta - \sin^2 \theta_c} \quad (5.38)$$

令 $k_3'' = i\kappa$

$$\kappa = k_1 \sqrt{\sin^2 \theta - \sin^2 \theta_c} \quad (5.39)$$

则折射波的电场为

$$\mathbf{E}'' = \mathbf{E}_0'' e^{-\kappa z} e^{i(k_1' x - \omega t)} \quad (5.40)$$

透入介质 2 的厚度与波长同数量级, 约为

$$\kappa^{-1} = \frac{\lambda_1}{2\pi \sqrt{\sin^2 \theta - \sin^2 \theta_c}} \quad (5.41)$$

其平均能流密度只有 x 分量, 无 z 分量.

$$\frac{E'}{E} = e^{-2i\varphi} \quad \tan \varphi = \frac{\sin^2 \theta - \sin^2 \theta_c}{\cos \theta} \quad (5.42)$$

5.4.6 反射系数与透射系数

Def. 11 反射系数, 透射系数

沿界面法线方向反射与折射的能流分量和入射能流分量之比, 称为反射系数 R 与透射系数 T .

$$\begin{cases} R = \frac{E_0'^2}{E_0^2} \\ T = \frac{Z_2^{-1} E_0''^2 \cos \theta''}{Z_1^{-1} E_0^2 \cos \theta} \end{cases} \quad (5.43)$$

Prop. 11 p 偏振与 s 偏振电磁波

$$\begin{cases} R_p = \left(\frac{Z_1 \cos \theta - Z_2 \cos \theta''}{Z_1 \cos \theta + Z_2 \cos \theta''} \right)^2 & R_s = \left(\frac{Z_2 \cos \theta - Z_1 \cos \theta''}{Z_2 \cos \theta + Z_1 \cos \theta''} \right)^2 \\ T_p = \frac{4Z_1 Z_2 \cos \theta \cos \theta''}{(Z_1 \cos \theta + Z_2 \cos \theta'')^2} & T_s = \frac{4Z_1 Z_2 \cos \theta \cos \theta''}{(Z_2 \cos \theta + Z_1 \cos \theta'')^2} \end{cases} \quad (5.44)$$

5.5 色散

5.5.1 色散介质的本构关系

Thm. 7 色散的本质

色散性质源于介质对外场的非瞬时响应. 对于介电常数

$$\mathbf{D}(\mathbf{x}) = \varepsilon(\omega) \mathbf{E}(\mathbf{x}) \quad \varepsilon(\omega) = - \int_{-\infty}^{\infty} \varepsilon(\tau) e^{i\omega\tau} d\tau \quad (5.45)$$

Pf

$$\begin{cases} \varepsilon(\mathbf{x}, \mathbf{x}', t, t') \\ \mu^{-1}(\mathbf{x}, \mathbf{x}', t, t') \end{cases} \xrightarrow[\text{平移不变性}]{\text{空间与时间的}} \begin{cases} \varepsilon(\mathbf{x} - \mathbf{x}', t - t') \\ \mu^{-1}(\mathbf{x} - \mathbf{x}', t - t') \end{cases}$$

$$\begin{cases} \mathbf{D}(\mathbf{x}, t) = \int \varepsilon(\mathbf{x} - \mathbf{x}', t - t') \mathbf{E}(\mathbf{x}', t') d^3x' dt' \\ \mathbf{H}(\mathbf{x}, t) = \int \mu^{-1}(\mathbf{x} - \mathbf{x}', t - t') \mathbf{B}(\mathbf{x}', t') d^3x' dt' \end{cases} \quad \begin{matrix} (t > t') \\ \text{因果关系} \end{matrix}$$

以电场为例, 假设 $\mathbf{E}(\mathbf{x}, t) = \mathbf{E}(\mathbf{x}) e^{-i\omega t}$:

$$\begin{aligned} \mathbf{D}(t, \mathbf{x}) &= \mathbf{E}(\mathbf{x}) \int_{-\infty}^t \varepsilon(t - t') e^{-i\omega t'} dt' = \mathbf{E}(\mathbf{x}) e^{-i\omega t} \int_{-\infty}^t \varepsilon(t - t') e^{i\omega(t-t')} dt' \\ &\stackrel{\tau=t-t'}{=} \mathbf{E}(\mathbf{x}) e^{-i\omega t} \int_0^{\infty} \varepsilon(\tau) e^{i\omega\tau} d\tau \stackrel{\varepsilon(\tau)=0, \tau<0}{=} \varepsilon(\omega) \mathbf{E}(\mathbf{x}) e^{-i\omega t} = \mathbf{D}(\mathbf{x}) e^{-i\omega t} \end{aligned}$$

Cor. 3

若磁导率与电导率存在色散, 有线性响应关系

$$\mathbf{H}(\mathbf{x}) = \mu^{-1}(\omega) \mathbf{B}(\mathbf{x}) \quad (5.46)$$

$$\mathbf{J}(\mathbf{x}) = \sigma(\omega) \mathbf{E}(\mathbf{x}) \quad (5.47)$$

5.5.2 Kramers-Kronig 关系

Thm. 8 Kramers-Kronig 关系

$$\operatorname{Re}\{\varepsilon_r(\omega)\} = 1 + \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\operatorname{Im}\{\varepsilon_r(\omega')\}}{\omega' - \omega} d\omega' \quad (5.48)$$

$$\operatorname{Im}\{\varepsilon_r(\omega)\} = \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\operatorname{Re}\{\varepsilon_r(\omega') - 1\}}{\omega' - \omega} d\omega' \quad (5.49)$$

5.5.3 一维波包在色散介质中的传播

Def. 12 波包

波包为一系列单色平面波的线性叠加, 其振幅为

$$\psi(t, x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k) e^{i[kx - \omega(k)t]} dk \quad (5.50)$$

其中 $A(k)$ 可由任意时刻 t_0 的振幅作 Fourier 逆变换得到

$$A(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(t_0, x) e^{-i\omega(k)t_0} e^{-ikx} dx \quad (5.51)$$

Lem. 2 位置-波数不确定性关系

$$\Delta k \cdot \Delta x \geq \frac{1}{2} \quad (5.52)$$

Prop. 12 波包的群速度与相速度

对于 $n = n(\omega)$ 的色散介质, 相速度为

$$v_p = \frac{\omega}{k} = \frac{c}{n(\omega)} \quad (5.53)$$

群速度为

$$v_g = \frac{d\omega}{dk} = \frac{c}{\frac{d}{d\omega}[\omega n(\omega)]} = \frac{c}{n(\omega) + \omega \frac{dn}{d\omega}} = \frac{v_p}{1 + \frac{\omega}{n} \frac{dn}{d\omega}} \quad (5.54)$$

因此只要 $\frac{dn}{d\omega} \neq 0$, 辄 $v_g \neq v_p$.

Prop. 13 线性色散

线性色散要求 $A(k)$ 在 $k = k_0$ 附近有一个较小的延展 Δk , 且 $\omega(k)$ 变化缓慢, 则有

$$\omega(k) \approx \omega(k_0) + \left. \frac{d\omega}{dk} \right|_{k=k_0} (k - k_0) = \omega_0 + v_g(k - k_0) \quad (5.55)$$

则波包的时间演化近似为

$$\psi(t, x) = \phi(0, x - v_g t) e^{i(k_0 v_g - \omega_0)t} \quad (5.56)$$

线性色散近似下, 波包的包络形状在传播过程中不变, 仅以 v_g 的速率整体平移.

Pf $\varphi = kx - \omega(k)t \approx kx - \omega_0 t - v_g k t + v_g k_0 t = k(x - v_g t) + (k_0 v_g - \omega_0)t$.

$$\psi(t, x) = \frac{1}{\sqrt{2\pi}} e^{i(k_0 v_g - \omega_0)t} \int_{-\infty}^{\infty} A(k) e^{ik(x - v_g t)} dk = \phi(0, x - v_g t) e^{i(k_0 v_g - \omega_0)t}.$$

5.5.4 非磁性绝缘介质的色散性质

Prop. 14 平面电磁波在色散介质中的 Maxwell 方程组

若非磁性的 ($\mu = \mu_0$) 各向同性线性介质的介电常量存在色散, 即 $\varepsilon = \varepsilon(\omega)$, 则平面电磁波在其中的 Maxwell 方程组可写为.

$$\begin{cases} \mathbf{k} \times \mathbf{E}_0 = \omega \mathbf{B}_0 \\ \mathbf{k} \times \mathbf{B}_0 = -\omega \mu_0 \varepsilon(\omega) \mathbf{E}_0 \\ \mathbf{k} \cdot \mathbf{E}_0 = 0 \\ \mathbf{k} \cdot \mathbf{B}_0 = 0 \end{cases} \quad (5.57)$$

Pf 将电磁波场量 $\mathbf{E}(t, \mathbf{x}) = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$, $\mathbf{B}(t, \mathbf{x}) = \mathbf{B}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$ 代入无源色散介质的 Maxwell 方程可得, 其中磁场旋度式为 $\nabla \times \mathbf{B} = \mu_0 \varepsilon(\omega) \frac{\partial \mathbf{E}}{\partial t}$.

Prop. 15 非磁性绝缘介质中的色散关系

$$k = k_0 \sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0}} \quad (5.58)$$

其中 $k_0 = \frac{\omega}{c}$ 为真空中波矢.

Pf 由 $\mathbf{k} \cdot \mathbf{E}_0 = 0$, 可得 $\mathbf{k} \times (\mathbf{k} \times \mathbf{E}_0) = \omega \mathbf{k} \times \mathbf{B}_0 = -\underbrace{\omega^2 \mu_0 \varepsilon(\omega)}_{=k^2} \mathbf{E}_0$.

$$\Rightarrow k = \omega \sqrt{\mu_0 \varepsilon(\omega)} = \frac{\omega}{c} \sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0}} = k_0 \sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0}}.$$

5.5.5 非磁性导电介质的色散性质

Prop. 16 导电介质的本构关系

设导电介质的介电常量存在色散, 则有

$$\mathbf{D}(\mathbf{x}) = \varepsilon_b(\omega) \mathbf{E}(\mathbf{x}) \quad (5.59)$$

其中的传导电流满足 Ohm 定律

$$\mathbf{J}(\mathbf{x}) = \sigma_c \mathbf{E}(\mathbf{x}) \quad (5.60)$$

Def. 13 导电介质的等效介电常量

$$\varepsilon_{\text{eff}} = \varepsilon_b(\omega) + i \frac{\sigma_c}{\omega} \quad (5.61)$$

Thm. 9 平面电磁波在导电介质中的 Maxwell 方程组

$$\begin{cases} \mathbf{k} \times \mathbf{E}_0 = \omega \mathbf{B}_0 \\ \mathbf{k} \times \mathbf{B}_0 = -\omega \mu_0 \left(\varepsilon_b + i \frac{\sigma_c}{\omega} \right) \mathbf{E}_0 = -\omega \mu_0 \varepsilon_{\text{eff}} \mathbf{E}_0 \\ \mathbf{k} \cdot \mathbf{E}_0 = 0 \\ \mathbf{k} \cdot \mathbf{B}_0 = 0 \end{cases} \quad (5.62)$$

Pf 导电介质中磁场旋度为 $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \mu_0 \sigma_c \mathbf{E} + \mu_0 \varepsilon_b(\omega) \frac{\partial \mathbf{E}}{\partial t}$, 将平面电磁波场量代入得证.

Prop. 17 非磁性导电介质中的色散关系

$$k = k_0 \sqrt{\frac{\varepsilon_{\text{eff}}}{\varepsilon_0}} \quad (5.63)$$

5.5.6 色散介质的穿透深度 (趋肤深度)

Lem. 3 复介电常量与复波矢量

色散介质中介电常量与波矢量均为复值, 可显式写出实部与虚部

$$\varepsilon(\omega) = \varepsilon_0 (\varepsilon'_r + i \varepsilon''_r) \quad (5.64)$$

$$\mathbf{k}(\omega) = \beta(\omega) + i \alpha(\omega) \quad (5.65)$$

可得 α 与 β 的关系

$$\begin{cases} \beta^2 - \alpha^2 = k_0^2 \varepsilon'_r \\ \beta \cdot \alpha = \frac{1}{2} k_0^2 \varepsilon''_r \end{cases} \quad (5.66)$$

一般 α 与 β 方向不同. 若相同, 则可解得

$$\begin{cases} \beta = k_0 \sqrt{\frac{\varepsilon_r}{2}} \sqrt{\sqrt{1 + \left(\frac{\varepsilon''_r}{\varepsilon'_r}\right)^2} + 1} \\ \alpha = k_0 \sqrt{\frac{\varepsilon_r}{2}} \sqrt{\sqrt{1 + \left(\frac{\varepsilon''_r}{\varepsilon'_r}\right)^2} - 1} \end{cases} \quad (5.67)$$

Pf 方向相同时 $\beta \cdot \alpha = \beta\alpha$, $\Rightarrow \frac{k_0^2 \varepsilon''_r}{2\beta} \Rightarrow \beta^4 - k_0^2 \varepsilon'_r \beta^2 - \frac{1}{4} k_0^4 \varepsilon''_r{}^2 = 0$. 利用求根公式可得 β , 反代得 α .

Prop. 18 色散介质中的电矢量

$$\mathbf{E}(t, \mathbf{x}) = \mathbf{E}_0 e^{-\alpha \cdot \mathbf{x}} e^{i(\beta \cdot \mathbf{x} - \omega t)} \quad (5.68)$$

Rmk 其中 $e^{i(\beta \cdot \mathbf{x} - \omega t)}$ 刻画电矢量的振荡, $e^{-\alpha \cdot \mathbf{x}}$ 刻画电矢量的衰减.

Pf $e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} = e^{i[(\beta + i\alpha) \cdot \mathbf{x} - \omega t]} = e^{-\alpha \cdot \mathbf{x} + i(\beta \cdot \mathbf{x} - \omega t)}$.

Def. 14 穿透深度

电磁波垂直入射后, 振幅衰减为 e^{-1} 倍时传播的距离, 称为该介质的穿透深度.

$$\delta = \frac{1}{\alpha} \quad (5.69)$$

Prop. 19 导电介质中的穿透深度

对于导电介质, $\varepsilon'_r = \frac{\varepsilon_b(\omega)}{\varepsilon_0}$, $\varepsilon''_r = \frac{\sigma_c}{\omega \varepsilon_0}$, 于是有

$$\begin{cases} \beta^2 - \alpha^2 = k_0^2 \frac{\varepsilon_b}{\varepsilon_0} = \omega^2 \mu_0 \varepsilon_b \\ \beta \cdot \alpha = \frac{1}{2} \omega \mu_0 \sigma_c \end{cases} \quad (5.70)$$

解得

$$\begin{cases} \beta = k_0 \sqrt{\frac{\varepsilon_b}{2\varepsilon_0}} \sqrt{\sqrt{1 + \left(\frac{\sigma_c}{\omega\varepsilon_b}\right)^2} + 1} \\ \alpha = k_0 \sqrt{\frac{\varepsilon_b}{2\varepsilon_0}} \sqrt{\sqrt{1 + \left(\frac{\sigma_c}{\omega\varepsilon_b}\right)^2} - 1} \end{cases} \quad (5.71)$$

Cor. 4 弱导电介质的近似

弱导电介质的电导率相对较小, 有 $\frac{\sigma_c}{\omega\varepsilon_b} \ll 1$, 近似得

$$\begin{cases} \beta \approx k_0 \sqrt{\frac{\varepsilon_b}{\varepsilon_0}} \\ \alpha \approx \frac{\sigma_c}{2} \sqrt{\frac{\mu_0}{\varepsilon_b}} \end{cases} \quad (5.72)$$

穿透深度为

$$\delta = \frac{2}{\sigma_c} \sqrt{\frac{\varepsilon_b}{\mu_0}} \quad (5.73)$$

Cor. 5 良导电介质的近似

良导电介质的电导率相对较大, 有 $\frac{\sigma_c}{\omega\varepsilon_b} \gg 1$, 近似得

$$\alpha \approx \beta \approx k_0 \sqrt{\frac{\sigma_c}{2\omega\varepsilon_0}} = \sqrt{\frac{\omega\mu_0\sigma_c}{2}} \quad (5.74)$$

穿透深度为

$$\delta = \sqrt{\frac{2}{\omega\mu_0\sigma_c}} \quad (5.75)$$

5.5.7 Lorenz 模型

Lorenz 模型是描述电介质的经典色散理论, 核心是束缚电子的受迫阻尼谐振

$$\varepsilon_r(\omega) = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - i\gamma\omega} \quad (5.76)$$

$$\operatorname{Re}\{\varepsilon_r(\omega)\} = 1 + \frac{\omega_p^2(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} \quad \operatorname{Im}\{\varepsilon_r(\omega)\} = \frac{\omega_p^2\gamma\omega}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} \quad (5.77)$$

其中 ω_p 为介质的等离子体频率, ω_0 为谐振本征频率, γ 为阻尼系数.

设束缚电子所受阻尼 $\mathbf{F}_{\text{damp}} = -m\gamma\dot{\mathbf{x}}$, 受电场力 $\mathbf{F}_e = -e\mathbf{E}_0 e^{-i\omega t}$, 运动方程为

$$m\ddot{\mathbf{x}} + m\omega_0^2 \mathbf{x} = -m\gamma\dot{\mathbf{x}} - e\mathbf{E}_0 e^{-i\omega t} \xrightarrow{\mathbf{x}(t)=\mathbf{x}_0 e^{-i\omega t}} \mathbf{x} = \frac{-e\mathbf{E}}{m(\omega_0^2 - \omega^2 - i\gamma\omega)}$$

$$\mathbf{P} = -n_e e \mathbf{x} = \frac{n_e e^2}{m} \frac{\mathbf{E}}{\omega_0^2 - \omega^2 - i\gamma\omega} \xrightarrow{\mathbf{P}=\epsilon_0 \chi \mathbf{E}} \chi(\omega) = \frac{\omega_p^2}{\omega_0^2 - \omega^2 - i\gamma\omega}$$

$$\epsilon_r(\omega) = 1 + \chi(\omega) = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - i\gamma\omega} = 1 + \frac{\omega_p^2(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2) + \gamma^2 \omega^2} + i \frac{\omega_p^2 \gamma \omega}{(\omega_0^2 - \omega^2) + \gamma^2 \omega^2}$$

5.5.8 Drude 模型

Drude 模型是描述金属导体的经典色散理论, 核心是带电粒子气模型近似. 设弛豫时间为 τ , 电导率为

$$\sigma(\omega) = \frac{\epsilon_0 \omega_p^2}{\tau^{-1} - i\omega} \xrightarrow[\text{长波近似}]{\omega \rightarrow 0} \sigma_c = \frac{n_e e^2}{m} \tau \quad (5.78)$$

等效相对介电常数为

$$\epsilon_r(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\tau^{-1}} \quad (5.79)$$

$$\text{Re}\{\epsilon_r\} = 1 - \frac{\omega_p^2 \tau^2}{1 + \omega^2 \tau^2} \quad \text{Im}\{\epsilon_r\} = \frac{\omega_p^2 \tau}{\omega(1 + \omega^2 \tau^2)} \quad (5.80)$$

设自由电子受散射力 $\mathbf{F}_{\text{sca}} = -\frac{m\mathbf{v}}{\tau}$, 受电场力 $\mathbf{F}_e = -e\mathbf{E}_0 e^{-i\omega t}$, 运动方程为

$$m\dot{\mathbf{v}} = -m\tau^{-1}\mathbf{v} - e\mathbf{E}_0 e^{-i\omega t} \xrightarrow{\mathbf{v}(t)=\mathbf{v}_0 e^{-i\omega t}} \mathbf{v} = \frac{-e\mathbf{E}}{m(\tau^{-1} - i\omega)}$$

$$\mathbf{J}_f = \rho \mathbf{v} = \frac{n_e e^2}{m} \frac{\mathbf{E}}{\tau^{-1} - i\omega} \xrightarrow{\mathbf{J}_f = \sigma \mathbf{E}} \sigma(\omega) = \frac{n_e e^2}{m} \frac{1}{\tau^{-1} - i\omega} = \frac{\epsilon_0 \omega_p^2}{\tau^{-1} - i\omega}$$

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = (\sigma(\omega) - i\omega\epsilon_0) \mathbf{E} = -i\omega\epsilon_{\text{eff}}(\omega) \mathbf{E} \implies \epsilon_{\text{eff}} = \epsilon_0 + \frac{i\sigma}{\omega}$$

5.6 电磁波在受限空间中的传播

5.6.1 波导的导模特征

Lem. 4 理想导体的边值关系

理想导体外表面满足

$$\begin{cases} E_t = B_n = 0 \\ \mathbf{n} \times \mathbf{H} = \mathbf{K} \\ \mathbf{n} \cdot \mathbf{D} = \sigma \end{cases} \quad (5.81)$$

在导体表面, 电场线与界面正交, 磁感线与界面相切.

Prop. 20

设波导管轴线方向为 z 方向, 则导模可表为

$$\begin{cases} \mathbf{E}(t, \mathbf{x}) = \mathbf{E}(x, y)e^{i(k_3 z - \omega t)} \\ \mathbf{H}(t, \mathbf{x}) = \mathbf{H}(x, y)e^{i(k_3 z - \omega t)} \end{cases} \quad (5.82)$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k^2 - k_3^2 \right) \begin{cases} \mathbf{E}(x, y) \\ \mathbf{H}(x, y) \end{cases} = 0 \quad (5.83)$$

Thm. 10 导模的横纵分解定理

(Transverse-Longitudinal Decomposition Theorem for Guided Modes)

电磁场的所有横向分量均可由纵向分量表示

$$\begin{cases} E_x = \frac{i}{k_t^2} \left(k_3 \frac{\partial E_z}{\partial x} + \omega \mu \frac{\partial H_z}{\partial y} \right) \\ E_y = \frac{i}{k_t^2} \left(k_3 \frac{\partial E_z}{\partial y} - \omega \mu \frac{\partial H_z}{\partial x} \right) \end{cases} \quad (5.84)$$

$$\begin{cases} H_x = \frac{i}{k_t^2} \left(-\omega \varepsilon \frac{\partial E_z}{\partial y} + k_3 \frac{\partial H_z}{\partial x} \right) \\ H_y = \frac{i}{k_t^2} \left(\omega \varepsilon \frac{\partial E_z}{\partial x} + k_3 \frac{\partial H_z}{\partial y} \right) \end{cases} \quad (5.85)$$

$$\text{Pf} \quad \frac{\partial}{\partial z} \rightarrow ik_3, \quad \frac{\partial}{\partial t} \rightarrow -i\omega. \quad \nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} = i\omega \mu \mathbf{H}, \quad \nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} = -i\omega \varepsilon \mathbf{E}.$$

$$\begin{cases} ik_3 E_y + i\omega \mu H_x = \frac{\partial E_z}{\partial y} \\ ik_3 E_x - i\omega \mu H_y = \frac{\partial E_z}{\partial x} \\ \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = i\omega \mu H_z \end{cases} \quad \begin{cases} i\omega \varepsilon E_x - ik_3 H_y = -\frac{\partial H_z}{\partial y} \\ i\omega \varepsilon E_y + ik_3 H_x = \frac{\partial H_z}{\partial x} \\ \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = -i\omega \varepsilon E_z \end{cases}$$

联立求解可得证.

Cor. 6

单边空心波导管不支持 TEM 模.

5.6.2 矩形波导管

Meth. 1 矩形波导管问题的求解

以 $u(t, x, y, z) = u(x, y)e^{i(k_3 z - \omega t)}$ 表示电磁场的任一直角坐标分量, 易得通解

$$u(x, y) = (C_1 \cos k_1 x + D_1 \sin k_1 x)(C_2 \cos k_2 y + D_2 \sin k_2 y) \quad (5.86)$$

结合 $x, y = 0$ 处边界条件, 可得

$$\begin{cases} E_x = A_1 \cos k_1 x \sin k_2 y e^{ik_3 x} \\ E_y = A_2 \sin k_1 x \cos k_2 y e^{ik_3 x} \\ E_z = A_3 \sin k_1 x \sin k_2 y e^{ik_3 x} \end{cases} \quad (5.87)$$

结合 $x = l_1, y = l_2$ 处边界条件, 可得

$$k_1 = \frac{n_1 \pi}{l_1} \quad k_2 = \frac{n_2 \pi}{l_2} \quad (n_1, n_2 = 0, 1, 2, \dots) \quad (5.88)$$

其中 n_1, n_2 不同时为零. 结合 $\nabla \cdot \mathbf{E} = 0$, 可得

$$k_1 A_1 + k_2 A_2 - ik_3 A_3 = 0 \quad (5.89)$$

再由 $\mathbf{H} = -\frac{i}{\omega \mu} \nabla \times \mathbf{E}$ 可得 $H_{x,y,z}$ 的取值

$$\begin{cases} H_x = -\frac{1}{\omega \mu} (k_3 A_2 + ik_2 A_3) \sin k_1 x \cos k_2 y e^{i(k_3 z - \omega t)} \\ H_y = \frac{1}{\omega \mu} (k_3 A_1 + ik_1 A_3) \cos k_1 x \sin k_2 y e^{i(k_3 z - \omega t)} \\ H_z = -\frac{i}{\omega \mu} (k_1 A_2 - k_2 A_1) \cos k_1 x \cos k_2 y e^{i(k_3 z - \omega t)} \end{cases} \quad (5.90)$$

Prop. 21 截止频率

要求 $k_3 = \sqrt{k^2 - k_1^2 - k_2^2}$ 为实值, 则有

$$k > k_{\text{cut}} = \pi \sqrt{\left(\frac{n_1}{l_1}\right)^2 + \left(\frac{n_2}{l_2}\right)^2} \quad (5.91)$$

截止角频率, 截止角频率, 截止波长分别为

$$\omega_{\text{cut}} = \frac{\pi}{\sqrt{\mu \varepsilon}} \sqrt{\left(\frac{n_1}{l_1}\right)^2 + \left(\frac{n_2}{l_2}\right)^2} \quad (5.92)$$

$$f_{\text{cut}} = \frac{1}{2\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{n_1}{l_1}\right)^2 + \left(\frac{n_2}{l_2}\right)^2} \quad (5.93)$$

$$\lambda_{\text{cut}} = \frac{2}{\sqrt{\left(\frac{n_1}{l_1}\right)^2 + \left(\frac{n_2}{l_2}\right)^2}} \quad (5.94)$$

若 $l_1 > l_2$, 则 TE_{10} 波有最低截止频率与截止波长

$$f_{\text{cut}} = \frac{v}{2l_1} \quad \lambda_{\text{cut}} = 2l_1 \quad (5.95)$$

5.6.3 谐振腔

Prop. 22 矩形谐振腔

设 $x = 0, l_1, y = 0, l_2, z = 0, l_3$ 为金属内腔壁, 腔内为各向同性的均匀线性介质, 介电常量为 ε , 磁导率为 μ . $u(x, y, z)$ 表示 $\mathbf{E}, \mathbf{H}$ 的任一 Cartesius 坐标分量, 满足 $\nabla^2 u + k^2 u = 0$, 其通解为

$$u(x, y, z) = (C_1 \cos k_1 x + D_1 \sin k_1 x)(C_2 \cos k_2 y + D_2 \sin k_2 y)(C_3 \cos k_3 z + D_3 \sin k_3 z) \quad (5.96)$$

结合边界条件与 $\nabla \cdot \mathbf{E} = 0$, 可得其特解为

$$\begin{cases} E_x = A_1 \cos k_1 x \sin k_2 y \sin k_3 z \\ E_y = A_2 \sin k_1 x \cos k_2 y \sin k_3 z \\ E_z = A_3 \sin k_1 x \sin k_2 y \cos k_3 z \end{cases} \quad (5.97)$$

$$k_1 = \frac{n_1 \pi}{l_1} \quad k_2 = \frac{n_2 \pi}{l_2} \quad k_3 = \frac{n_3 \pi}{l_3} \quad (n_1, n_2, n_3 = 0, 1, 2, \dots) \quad (5.98)$$

$$k_1 A_1 + k_2 A_2 + k_3 A_3 = 0 \quad (5.99)$$

谐振腔的共振频率为

$$\omega = \frac{\pi}{\sqrt{\mu\varepsilon}} \sqrt{\frac{n_1^2}{l_1^2} + \frac{n_2^2}{l_2^2} + \frac{n_3^2}{l_3^2}} \quad (5.100)$$

5.7 电磁辐射

5.7.1 电磁场的势

Def. 15 标势

$$-\nabla\phi(t, \mathbf{x}) = \mathbf{E}(t, \mathbf{x}) + \frac{\partial \mathbf{A}(t, \mathbf{x})}{\partial t} \quad (5.101)$$

Rmk 时变电场的势表述为

$$\mathbf{E}(t, \mathbf{x}) = -\nabla\phi(t, \mathbf{x}) - \frac{\partial \mathbf{A}(t, \mathbf{x})}{\partial t} \quad (5.102)$$

Pf 由 $\nabla \cdot \mathbf{B}(t, \mathbf{x})$, 引入矢势 $\mathbf{B}(t, \mathbf{x}) = \nabla \times \mathbf{A}(t, \mathbf{x})$. 于是 $\nabla \times \mathbf{E}(t, \mathbf{x}) = -\frac{\partial}{\partial t} \nabla \times \mathbf{A}(t, \mathbf{x})$, 从而 $\nabla \times \left[\underbrace{\mathbf{E}(t, \mathbf{x}) + \frac{\partial \mathbf{A}(t, \mathbf{x})}{\partial t}}_{=-\nabla\phi(t, \mathbf{x})} \right] = 0$.

Lem. 5 有源电磁场的波动方程

$$\begin{cases} \square \mathbf{E} = \frac{1}{\varepsilon_0} \nabla \rho + \mu_0 \frac{\partial \mathbf{J}}{\partial t} \\ \square \mathbf{B} = -\mu_0 \nabla \times \mathbf{J} \end{cases} \quad (5.103)$$

Prop. 23 电磁势的 Maxwell 方程组

$$\begin{cases} \nabla^2 \phi + \frac{\partial}{\partial t} \nabla \cdot \mathbf{A} = -\frac{\rho}{\varepsilon_0} \\ \square \mathbf{A} - \nabla \left(\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} \right) = -\mu_0 \mathbf{J} \end{cases} \quad (5.104)$$

Pf $\nabla \cdot \mathbf{E} = \nabla \cdot \left(-\nabla \phi - \frac{\partial \mathbf{A}}{\partial t} \right) = -\nabla^2 \phi - \frac{\partial}{\partial t} \nabla \cdot \mathbf{A} = \frac{\rho}{\varepsilon_0}, \Rightarrow \nabla^2 \phi + \frac{\partial}{\partial t} \nabla \cdot \mathbf{A} = -\frac{\rho}{\varepsilon_0}$.
 $\nabla \times \mathbf{B} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J} + \frac{1}{c^2} \frac{\partial}{\partial t} \left(-\nabla \phi - \frac{\partial \mathbf{A}}{\partial t} \right) = \mu_0 \mathbf{J} - \frac{1}{c^2} \nabla \frac{\partial \phi}{\partial t} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2}$,
 $\Rightarrow \nabla^2 \mathbf{A} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} - \nabla \left(\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} \right) = -\mu_0 \mathbf{J}$.

Prop. 24 Lorentz 力的电磁势表述

$$\frac{d\mathbf{p}_c}{dt} = \frac{d}{dt}(\mathbf{p} + q\mathbf{A}) = -q\nabla(\phi - \mathbf{v} \cdot \mathbf{A}) \quad (5.105)$$

Prop. 25 规范变换

$$\begin{cases} \mathbf{A} & \rightarrow \mathbf{A}' = \mathbf{A} + \nabla\psi \\ \phi & \rightarrow \phi' = \phi - \frac{\partial\psi}{\partial t} \end{cases} \quad (5.106)$$

Prop. 26

在 Coulomb 规范 $\nabla \cdot \mathbf{A}_C(t, \mathbf{x}) = 0$ 下, 方程化为

$$\begin{cases} \nabla^2 \phi_C(t, \mathbf{x}) = -\frac{\rho(t, \mathbf{x})}{\varepsilon_0} \\ \nabla^2 \mathbf{A}_C(t, \mathbf{x}) - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}_C(t, \mathbf{x})}{\partial t^2} - \frac{1}{c^2} \frac{\partial}{\partial t} \nabla \phi_C(t, \mathbf{x}) = -\mu_0 \mathbf{J}(t, \mathbf{x}) \end{cases} \quad (5.107)$$

其中标势表示为

$$\phi_C(t, \mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(t, \mathbf{x})}{r} d^3x \quad (5.108)$$

5.7.2 Lorenz 规范与 d'Alembert 方程

Def. 16 Lorenz 规范

$$\nabla \cdot \mathbf{A}_L(t, \mathbf{x}) + \frac{1}{c^2} \frac{\partial \phi_L(t, \mathbf{x})}{\partial t} = 0 \quad (5.109)$$

Prop. 27 d'Alembert 方程

$$\begin{cases} \square \phi_L(t, \mathbf{x}) = -\frac{\rho(t, \mathbf{x})}{\varepsilon_0} \\ \square \mathbf{A}_L(t, \mathbf{x}) = -\mu_0 \mathbf{J}(t, \mathbf{x}) \end{cases} \quad (5.110)$$

Rmk 形式上, 电荷产生标势波动, 电流产生矢势波动.

$$\text{Pf} \quad \nabla^2 \phi_L + \frac{\partial}{\partial t} \nabla \cdot \mathbf{A}_L = \nabla^2 \phi_L - \frac{1}{c^2} \frac{\partial^2 \phi_L}{\partial t^2} = -\frac{\rho}{\varepsilon_0}. \quad \square \mathbf{A} - \underbrace{\nabla \left(\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} \right)}_{=0} = -\mu_0 \mathbf{J}.$$

5.7.3 推迟势

Lem. 6 三维无界波动问题的球对称通解

对于 $\square \phi = 0$, 若 $\rho(t, \mathbf{x}) = \phi(t, r)$, 则方程可化为

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = 0 \quad \implies \frac{\partial^2(r\phi)}{\partial r^2} - \frac{1}{c^2} \frac{\partial^2(r\phi)}{\partial t^2} = 0 \quad (5.111)$$

对于一维波动方程, 有行波解

$$u(t, r) = f_1(t - r/c) + f_2(t + r/c) \quad (5.112)$$

从而球面波解为

$$\phi(t, r) = \frac{1}{r} f_1(t - r/c) + \frac{1}{r} f_2(t + r/c) \quad (5.113)$$

Def. 17 推迟时刻

t 时刻的势由 $t - r/c$ 时刻的源决定, 定义此时为推迟时刻

$$t_{\text{rev}} = t - \frac{r}{c} \quad (5.114)$$

$t - t_{\text{rev}}$ 即为电磁相互作用从源点传播到场点所需的时间.

Prop. 28 原点处含时点电荷的推迟势

设原点处有一电荷量 $Q(t)$ 随时间变化的点电荷, 其产生的标势 $\phi(t, \mathbf{x})$ 的波动方程为

$$\square \phi(t, \mathbf{x}) = -\frac{1}{\varepsilon_0} Q(t) \delta(\mathbf{x}) \quad (5.115)$$

$r \neq 0$ 时化为齐次波动方程, 解为

$$\phi(t, r) = \frac{Q(t_{\text{rev}})}{4\pi\varepsilon_0 r} \quad (5.116)$$

$$\text{Pf } \nabla t_{\text{rev}} = -\frac{1}{c} \nabla r = -\frac{1}{c} \mathbf{e}_r.$$

$$\begin{aligned} \nabla^2 \frac{Q(t_{\text{rev}})}{r} &= Q(t_{\text{rev}}) \nabla^2 \frac{1}{r} + 2 \nabla \frac{1}{r} \cdot \nabla Q(t_{\text{rev}}) + \frac{1}{r} \nabla^2 Q(t_{\text{rev}}) \\ &= -4\pi \delta(\mathbf{x}) Q(t) + 2 \left(-\frac{1}{r^2} \right) \left(-\frac{1}{c} \frac{\partial Q}{\partial t} \right) + \frac{1}{r} \left(\frac{1}{c^2} \frac{\partial^2 Q}{\partial t^2} - \frac{2}{cr} \frac{\partial Q}{\partial t} \right) \\ &= \frac{1}{c^2 r} \frac{\partial^2 Q}{\partial t^2} - 4\pi Q(t) \delta(\mathbf{x}) = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \frac{Q}{r} - 4\pi Q(t) \delta(\mathbf{x}) \end{aligned}$$

两端除以 $4\pi\varepsilon_0$ 可得 $\square \phi(t, \mathbf{x}) = -Q(t) \delta(\mathbf{x}) / \varepsilon_0$.

Cor. 7 含时点电荷的标势

$\mathbf{x}'$ 处的点电荷在空间产生的标势为

$$\phi(t, \mathbf{x}) = \frac{Q(t_{\text{rev}}, \mathbf{x}')}{4\pi\varepsilon_0 r} \quad (5.117)$$

其中 $r = |\mathbf{x} - \mathbf{x}'|$.

Cor. 8 连续带电体的标势

根据叠加原理, 连续电荷分布 $\rho(t, \mathbf{x}')$ 在空间产生的标势为

$$\phi(t, \mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(t_{\text{rev}}, \mathbf{x}')}{r} d^3x' \quad (5.118)$$

Cor. 9 电流的矢势

含时电流分布 $\mathbf{J}(t, \mathbf{x}')$ 在空间产生的矢势为

$$\mathbf{A}(t, \mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(t_{\text{rev}}, \mathbf{x}')}{r} d^3x' \quad (5.119)$$

Rmk 易证, 上述 $\phi(t, \mathbf{x})$ 与 $\mathbf{A}(t, \mathbf{x})$ 满足 Lorenz 规范.

5.7.4 时谐电流的多级辐射**Lem. 7** 时谐辐射电磁场

已知辐射矢势分布 $\mathbf{A}(t, \mathbf{x})$, 即可求出辐射电磁场.

$$\mathbf{B}(t, \mathbf{x}) = \nabla \times \mathbf{A}(t, \mathbf{x}) \quad (5.120)$$

$$\mathbf{E}(t, \mathbf{x}) = \frac{ic^2}{\omega} \nabla \times \mathbf{B}(t, \mathbf{x}) \quad (5.121)$$

Prop. 29 时谐电流的推迟势

给定电流分布 $\mathbf{J}(t, \mathbf{x}') = \mathbf{J}(\mathbf{x}')e^{-i\omega t}$, 可得矢势

$$\mathbf{A}(t, \mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}')}{r} e^{i(kr - \omega t)} d^3x' = \mathbf{A}(\mathbf{x})e^{-i\omega t} \quad (5.122)$$

$$\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}')}{r} e^{ikr} d^3x' \quad (5.123)$$

由 Lorenz 规范可得标势

$$\phi(t, \mathbf{x}) = \phi(\mathbf{x})e^{-i\omega t} = \frac{c^2}{i\omega} \nabla \cdot \mathbf{A}(\mathbf{x})e^{-i\omega t} \quad (5.124)$$

Rmk 也可由电荷守恒定律求出电荷分布

$$\rho(t, \mathbf{x}') = \frac{1}{i\omega} \nabla' \cdot \mathbf{J}(t, \mathbf{x}') \quad (5.125)$$

再利用连续带电体的推迟势公式求出 $\phi(t, \mathbf{x})$.

Prop. 30 小区域电流产生的辐射

小区域分布要求 $l \ll r, l \ll \lambda$, 其中 l 为电流分布区域的线度, $r = |\mathbf{x} - \mathbf{x}'|$, λ 为辐射波长.

1. $r \ll \lambda$ 为近场区, $e^{ikr} \sim 1$, $\mathbf{A}(t, \mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}')}{r} d^3x'$, 电磁场的空间依赖形式与静电场相似.
2. $r \sim \lambda$ 为感应区.
3. $r \gg \lambda$ 为远场区.

Rmk 已知远场区矢势 $\mathbf{A}(t, \mathbf{x})$, 则电磁场为

$$\mathbf{B}(t, \mathbf{x}) = \nabla \times \mathbf{A}(t, \mathbf{x}) = ik\mathbf{e}_R \times \mathbf{A}(t, \mathbf{x}) \quad (5.126)$$

$$\mathbf{E}(t, \mathbf{x}) = c\mathbf{B}(t, \mathbf{x}) \times \mathbf{e}_R \quad (5.127)$$

其中 $R = |\mathbf{x}|$, $\mathbf{e}_R = \frac{\mathbf{x}}{R}$.

Prop. 31 时谐电流推迟势的多极展开

令 $R = |\mathbf{x}|$, 在 $r = |\mathbf{x} - \mathbf{x}'| \gg |\mathbf{x}'|$ 处, 有 $r \approx R - \mathbf{e}_R \cdot \mathbf{x}'$.

$$\begin{aligned} \mathbf{A}(\mathbf{x}) &= \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}') e^{ik(R - \mathbf{e}_R \cdot \mathbf{x}')}}{R - \mathbf{e}_R \cdot \mathbf{x}'} d^3x' = \frac{\mu_0 e^{ikR}}{4\pi R} \int \mathbf{J}(\mathbf{x}') e^{-ik\mathbf{e}_R \cdot \mathbf{x}'} d^3x' \\ &= \frac{\mu_0 e^{ikR}}{4\pi R} \int \mathbf{J}(\mathbf{x}') (1 - ik\mathbf{e}_R \cdot \mathbf{x}' + \dots) d^3x' = \mathbf{A}^{(0)}(\mathbf{x}) + \mathbf{A}^{(1)}(\mathbf{x}) + \dots \quad (5.128) \\ &= \frac{\mu_0 e^{ikR}}{4\pi R} \int \mathbf{J}(\mathbf{x}') d^3x' - \frac{ik\mu_0 e^{ikR}}{4\pi R} \int \mathbf{J}(\mathbf{x}') (\mathbf{e}_R \cdot \mathbf{x}') d^3x' + \dots \end{aligned}$$

5.7.5 电偶极辐射

Lem. 8

若在边界处 $\mathbf{n} \cdot \mathbf{J}(\mathbf{x}) = 0$, 则有

$$\int \mathbf{J}(\mathbf{x}) d^3x = - \int \nabla \cdot \mathbf{J}(\mathbf{x}) \mathbf{x} d^3x \quad (5.129)$$

Pf 据 $\int \nabla \cdot \mathbf{T} = \oint \mathbf{n} \cdot \mathbf{T} d^2x$, 有

$$\begin{aligned} I &= \int \nabla \cdot [\mathbf{J}(\mathbf{x}) \mathbf{x}] d^3x = \int [\nabla \cdot \mathbf{J}(\mathbf{x})] \mathbf{x} d^3x + \int \mathbf{J}(\mathbf{x}) \cdot \nabla \mathbf{x} d^3x \\ &= \int \nabla \cdot \mathbf{J}(\mathbf{x}) \mathbf{x} d^3x + \int \mathbf{J}(\mathbf{x}) d^3x = \oint \underbrace{\mathbf{n} \cdot \mathbf{J}(\mathbf{x}) \mathbf{x}}_{=0} d^2x = 0 \end{aligned}$$

Def. 18 时谐电流体系的电偶极矩

$$\mathbf{p}(t) = \mathbf{p}e^{-i\omega t} = e^{-i\omega t} \int \rho(\mathbf{x}') \mathbf{x}' d^3x' = e^{-i\omega t} \frac{i}{\omega} \int \mathbf{J}(\mathbf{x}') d^3x' \quad (5.130)$$

Rmk

$$\dot{\mathbf{p}} = \frac{d\mathbf{p}(t)}{dt} = e^{-i\omega t} \int \mathbf{J}(\mathbf{x}') d^3x' \quad (5.131)$$

$$\text{Pf} \quad \int \rho(\mathbf{x}') \mathbf{x}' d^3x' = \frac{1}{i\omega} \int \nabla' \cdot \mathbf{J}(\mathbf{x}') \mathbf{x}' d^3x' = \frac{i}{\omega} \int \mathbf{J}(\mathbf{x}') d^3x'.$$

Prop. 32 电偶极子的辐射矢势

$$\mathbf{A}^{(0)}(t, \mathbf{x}) = \frac{\mu_0 e^{ikR}}{4\pi R} \dot{\mathbf{p}}(t) \quad (5.132)$$

$$\text{Rmk} \quad \text{由 } \nabla \frac{e^{ikR}}{R} = \left(-\frac{\mathbf{e}_R}{R^2} + \frac{ik\mathbf{e}_R}{R} \right) e^{ikR} \approx ik\mathbf{e}_R \frac{e^{ikR}}{R} \text{ 可得}$$

$$\nabla \rightarrow ik\mathbf{e}_R \quad (5.133)$$

Prop. 33 电偶极子的辐射场

$$\mathbf{B}^{(0)}(t, \mathbf{x}) = \frac{\mu_0 e^{ikR}}{4\pi R} ik\mathbf{e}_R \times \dot{\mathbf{p}}(t) = \frac{\mu_0 e^{ikR}}{4\pi cR} \ddot{\mathbf{p}}(t) \times \mathbf{e}_R \quad (5.134)$$

$$\mathbf{E}^{(0)}(t, \mathbf{x}) = \frac{\mu_0 e^{ikR}}{4\pi R} (\ddot{\mathbf{p}} \times \mathbf{e}_R) \times \mathbf{e}_R \quad (5.135)$$

Lem. 9

设球坐标的极轴为 z 轴, 则有

$$\mathbf{e}_z \times \mathbf{e}_R = \mathbf{e}_\varphi \sin \theta \quad (5.136)$$

Prop. 34 球坐标中的辐射场

设电偶极矩沿极轴 z 方向振荡 $\mathbf{p} = \mathbf{e}_z p$

$$\mathbf{B}^{(0)}(t, \mathbf{x}) = -\frac{\mu_0 \omega^2}{4\pi cR} e^{i(kR - \omega t)} p \sin \theta \mathbf{e}_\varphi = \frac{\mu_0 e^{ikR}}{4\pi cR} \ddot{p} \sin \theta \mathbf{e}_\varphi \quad (5.137)$$

由 $\mathbf{e}_\varphi \times \mathbf{e}_R = \mathbf{e}_\theta$ 可得

$$\mathbf{E}^{(0)}(t, \mathbf{x}) = -c\mathbf{e}_R \times \mathbf{B}^{(0)}(t, \mathbf{x}) = -\frac{\mu_0 \omega^2}{4\pi R} e^{i(kR - \omega t)} p \sin \theta \mathbf{e}_\theta = \frac{\mu_0 e^{ikR}}{4\pi R} \ddot{p} \sin \theta \mathbf{e}_\theta \quad (5.138)$$

Prop. 35 辐射能流

时谐电偶极子辐射场的 Poynting 矢量为

$$\begin{aligned} S(t, R, \theta) &= \frac{1}{\mu_0} \operatorname{Re}\{\mathbf{E}^{(0)}(t, R, \theta)\} \times \operatorname{Re}\{\mathbf{B}^{(0)}(t, R, \theta)\} \\ &= \frac{\mu_0 \omega^4 p^2}{16\pi^2 c R^2} \sin^2 \theta \cos^2(kR - \omega t) \mathbf{e}_R \end{aligned} \quad (5.139)$$

$$\langle \mathbf{S} \rangle(R, \theta) = \frac{\mu_0 \omega^4 p^2}{32\pi^2 c R^2} \sin^2 \theta \mathbf{e}_R \quad (5.140)$$

Rmk 辐射能流密度的角分布函数为

$$f(\theta) = \sin^2 \theta \quad (5.141)$$

Prop. 36 辐射功率

$$P = \frac{\mu_0}{12\pi c} \omega^4 p^2 \quad (5.142)$$

Pf $P = \oint \langle S \rangle d^2x, I = \int_{\theta=0}^{\pi} \int_{\varphi=0}^{2\pi} \frac{\sin^2 \theta}{R^2} R^2 \sin \theta d\theta d\varphi = 2\pi \int_0^{\pi} \sin^3 \theta d\theta = \frac{8\pi}{3}.$

Cor. 10 时谐电偶极子远场区辐射场的特征

1. 辐射电磁波近似为 TEM 波;
2. 场量振幅 $E, B \propto \frac{1}{R}$;
3. 在 4π 的立体角内, 总辐射功率守恒, 与距离 R 无关;
4. 能流密度与辐射功率 $S, P \propto \omega^4$.

Eg. 1 中心馈电细短直天线的远场辐射

设天线长度 $l \ll \lambda$, 电流为

$$I(t, z) = I_0 \left(1 - \frac{2}{l}|z|\right) e^{-\omega t} \quad (|z| \leq l/2) \quad (5.143)$$

电偶极矩为

$$\mathbf{p} = \frac{i}{\omega} \int_{-l/2}^{l/2} I(z) dz \mathbf{e}_z = \frac{i}{2\omega} I_0 l \mathbf{e}_z \quad (5.144)$$

辐射功率为

$$P = \frac{\pi}{12} Z_0 I_0^2 \left(\frac{l}{\lambda}\right)^2 \quad (5.145)$$

定义等效电阻 R_r , 满足 $P = \frac{1}{2}I_0^2 R_r$. 设 $\kappa = l/\lambda$, 则有

$$R_r = \frac{\pi}{6} Z_0 \left(\frac{l}{\lambda} \right)^2 = \kappa^2 197 \Omega \quad (5.146)$$

Eg. 2 无辐射体系

1. 稳恒源: $\frac{\partial \rho}{\partial t} = \frac{\partial \mathbf{J}}{\partial t} = 0$, 电磁场不随时间变化, 因此无辐射.
2. 球对称径向运动: 球对称性使所有电磁多极矩均为零, 因此无辐射.

5.7.6 磁偶极辐射与电四极辐射

Def. 19 时谐电流体系的磁偶极矩与电四极矩

$$\mathbf{m} = \frac{1}{2} \int \mathbf{x}' \times \mathbf{J}(\mathbf{x}') d^3 x' \quad (5.147)$$

$$\mathcal{D} = \frac{1}{2} \int \rho(\mathbf{x}') \mathbf{x}' \mathbf{x}' d^3 x' \quad (5.148)$$

Lem. 10

$$\int \mathbf{J} \mathbf{x} + \mathbf{x} \mathbf{J} d^3 x = - \int (\nabla \cdot \mathbf{J}) \mathbf{x} \mathbf{x} d^3 x \quad (5.149)$$

$$\text{Pf } \nabla \cdot (\mathbf{J} \mathbf{x} \mathbf{x}) = (\nabla \cdot \mathbf{J}) \mathbf{x} \mathbf{x} + \underbrace{\mathbf{J} \cdot \nabla \mathbf{x}}_{=\mathcal{I}} \mathbf{x} + \mathbf{x} \underbrace{\mathbf{J} \cdot \nabla \mathbf{x}}_{=\mathcal{I}} = (\nabla \cdot \mathbf{J}) \mathbf{x} \mathbf{x} + \mathbf{J} \mathbf{x} + \mathbf{x} \mathbf{J}.$$

Prop. 37 磁偶极辐射与电四极辐射的矢势

$\mathbf{A}^{(1)}(\mathbf{x})$ 可利用二阶张量的对称化与反对称化分为两项

$$\mathbf{A}^{(1)}(\mathbf{x}) = -ik \frac{\mu_0 e^{ikR}}{4\pi R} \mathbf{e}_R \cdot \int \mathbf{x}' \mathbf{J} d^3 x = -ik \frac{\mu_0 e^{ikR}}{4\pi R} \mathbf{e}_R \cdot \int \underbrace{\frac{1}{2}(\mathbf{x}' \mathbf{J} - \mathbf{J} \mathbf{x}')}_{\text{AS}} + \underbrace{\frac{1}{2}(\mathbf{x}' \mathbf{J} + \mathbf{J} \mathbf{x}')}_{\text{S}} d^3 x \quad (5.150)$$

反对称部分对应磁偶极辐射场的矢势

$$\mathbf{A}_m^{(1)}(t, \mathbf{x}) = \mathbf{A}_{\text{AS}}^{(1)}(\mathbf{x}) e^{-i\omega t} = ik \frac{\mu_0 e^{ikR}}{4\pi R} \mathbf{e}_R \times \mathbf{m}(t) \quad (5.151)$$

对称部分对应电四极辐射场的矢势

$$\mathbf{A}_D^{(1)}(t, \mathbf{x}) = \mathbf{A}_S^{(1)}(\mathbf{x}) e^{-i\omega t} = ik \frac{\mu_0 e^{ikR}}{4\pi R} [i\omega \mathbf{e}_R \cdot \mathcal{D}(t)] = \frac{\mu_0 e^{ikR}}{4\pi c R} [\mathbf{e}_R \cdot \ddot{\mathcal{D}}(t)] \quad (5.152)$$

于是可将矢势展开的第二极写为

$$\mathbf{A}^{(1)}(t, \mathbf{x}) = \mathbf{A}_m^{(1)}(t, \mathbf{x}) + \mathbf{A}_D^{(1)}(t, \mathbf{x}) = ik \frac{\mu_0 e^{ikR}}{4\pi R} (\mathbf{e}_R \times \mathbf{m} - \mathbf{e}_R \cdot \dot{\mathbf{D}}) \quad (5.153)$$

Rmk 交变电流情形有 $\nabla \cdot \mathbf{J} = i\omega\rho$, 因此电流体系可同时产生电多极辐射与磁多极辐射.

$$\begin{aligned} \text{Pf} \quad & \frac{1}{2} \int (\mathbf{e}_R \cdot \mathbf{x}') \mathbf{J} - (\mathbf{e}_R \cdot \mathbf{J}) \mathbf{x}' d^3x' = -\mathbf{e}_R \times \left(\frac{1}{2} \int \mathbf{x}' \times \mathbf{J} d^3x' \right) = -\mathbf{e}_R \times \mathbf{m}. \\ & \frac{1}{2} \int \mathbf{J} \mathbf{x}' \mathbf{x}' \mathbf{J} d^3x' = -\frac{1}{2} \int (\nabla' \cdot \mathbf{J}) \mathbf{x}' \mathbf{x}' d^3x' = -i\omega \frac{1}{2} \int \rho(\mathbf{x}') \mathbf{x}' \mathbf{x}' d^3x' = -i\omega \mathcal{D}. \end{aligned}$$

Prop. 38 磁偶极辐射场的场量

$$\mathbf{B}_m^{(1)}(t, \mathbf{x}) = \frac{\mu_0 e^{ikR}}{4\pi c^2 R} [\ddot{\mathbf{m}}(t) \times \mathbf{e}_R] \times \mathbf{e}_R = \frac{\mu_0 e^{ikR}}{4\pi c^2 R} \ddot{m} \sin \theta \mathbf{e}_\theta \quad (5.154)$$

$$\mathbf{E}_m^{(1)}(t, \mathbf{x}) = -\frac{\mu_0 e^{ikR}}{4\pi c R} \ddot{\mathbf{m}}(t) \times \mathbf{e}_R = -\frac{\mu_0 e^{ikR}}{4\pi c R} \ddot{m} \sin \theta \mathbf{e}_\varphi \quad (5.155)$$

Rmk 可由电偶极辐射场作代换得到磁偶极辐射场:

$$\begin{cases} \mathbf{p} \rightarrow \frac{\mathbf{m}}{c} \\ \mathbf{E} \rightarrow c\mathbf{B} \\ c\mathbf{B} \rightarrow -\mathbf{E} \end{cases} \quad (5.156)$$

Prop. 39 磁偶极辐射场的能流密度与辐射功率

$$\langle \mathbf{S}_m \rangle = \frac{\mu_0 \omega^4 m^2}{32\pi^2 c^3 R^2} \sin^2 \theta \mathbf{e}_R \quad (5.157)$$

$$P_m = \frac{\mu_0 \omega^4 m^2}{12\pi c^3} \quad (5.158)$$

Prop. 40 电四极辐射场的场量

$$\mathbf{B}_D^{(1)}(t, \mathbf{x}) = \frac{\mu_0 e^{ikR}}{4\pi c^2 R} [\mathbf{e}_R \times \ddot{\mathbf{D}}(t)] \times \mathbf{e}_R \quad (5.159)$$

$$\mathbf{E}_D^{(1)}(t, \mathbf{x}) = \frac{\mu_0 e^{ikR}}{4\pi c R} \{ [\mathbf{e}_R \cdot \ddot{\mathbf{D}}(t)] \nabla \times \mathbf{e}_R \} \mathbf{e}_R \quad (5.160)$$

Prop. 41 电四极辐射场的能流密度

$$\langle \mathbf{S}_D \rangle = \frac{1}{4\pi\epsilon_0} \frac{1}{8\pi c^5 R^2} |[\mathbf{e}_R \cdot \ddot{\mathbf{D}}(t)] \times \mathbf{e}_R|^2 \mathbf{e}_R \quad (5.161)$$

5.7.7 天线辐射

Prop. 42 细长直天线的电流形式

设天线沿 z 轴, 则天线表面电流为 $\mathbf{J} = J_z \mathbf{e}_z$, 矢势 $\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J} e^{ikr}}{r} d^3x' = A_z \mathbf{e}_z$.

Lorenz 规范化为

$$\frac{\partial A_z}{\partial z} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} = 0 \quad (5.162)$$

由天线表面电场 $E_z = 0$, 可得 A_z 满足的波动方程

$$\frac{\partial^2 A_z}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2 A_z}{\partial t^2} = 0 \quad (5.163)$$

结合电流边界条件 $\mathbf{J}(\mathbf{x}')|_{\text{endpoints}} = 0$, 从而 $\mathbf{J}(\mathbf{x}')$ 有驻波形式.

Prop. 43 对称细长直天线的辐射

对称细长直天线的电流分布为

$$I(t, z) = I_0 \sin k(l/2 - |z|) e^{-i\omega t} \quad (|z| \leq l/2) \quad (5.164)$$

其矢势分量 A_z 为

$$\begin{aligned} A_z(t, \mathbf{x}) &= \frac{\mu_0}{4\pi} \int_{-l/2}^{l/2} \frac{I(t_{\text{rev}}, z)}{r} dz = \frac{\mu_0 I_0}{4\pi} \int_{-l/2}^{l/2} \frac{\sin k(l/2 - |z|)}{r} e^{-i\omega(t-r/c)} dz \\ &\approx \frac{\mu_0 I_0}{4\pi R} e^{i(kR-\omega t)} \int_{-l/2}^{l/2} \sin \left(\frac{kl}{2} - l|z| \right) e^{-ikz \cos \theta} dz \\ &= \frac{\mu_0 I_0}{2\pi k R} e^{i(kR-\omega t)} \frac{1}{\sin^2 \theta} \left(\cos \frac{kl \cos \theta}{2} - \cos \frac{kl}{2} \right) \end{aligned} \quad (5.165)$$

其辐射电磁场为

$$\mathbf{B}(t, \mathbf{x}) = -i \frac{\mu_0 I_0}{2\pi R} e^{i(kR-\omega t)} \frac{1}{\sin \theta} \left(\cos \frac{kl \cos \theta}{2} - \cos \frac{kl}{2} \right) \mathbf{e}_\varphi \quad (5.166)$$

$$\mathbf{E}(t, \mathbf{x}) = -i \frac{\mu_0 I_0 c}{2\pi R} e^{i(kR-\omega t)} \frac{1}{\sin \theta} \left(\cos \frac{kl \cos \theta}{2} - \cos \frac{kl}{2} \right) \mathbf{e}_\theta \quad (5.167)$$

辐射能流密度为

$$\langle \mathbf{S} \rangle = \frac{\mu_0 I_0 c}{8\pi^2 R^2} \underbrace{\frac{1}{\sin^2 \theta} \left(\cos \frac{kl \cos \theta}{2} - \cos \frac{kl}{2} \right)^2}_{=f(\theta)} \mathbf{e}_R \quad (5.168)$$

Eg. 3 半波天线

取对称细长直天线长度为 $l = \nu \frac{\lambda}{2}$, 则称之为半波天线. 对于 $\nu = 1$ 的半波天线:

$$I(t, z) = I_0 \cos(kz) e^{-i\omega t} \quad (|z| \leq \lambda/4) \quad (5.169)$$

$$\mathbf{A}(t, \mathbf{x}) = \frac{\mu_0 I_0}{2\pi k R} e^{i(kR - \omega t)} \frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin^2 \theta} \mathbf{e}_z \quad (5.170)$$

$$\mathbf{B}(t, \mathbf{x}) = -i \frac{\mu_0 I_0}{2\pi R} e^{i(kR - \omega t)} \frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \mathbf{e}_\varphi \quad (5.171)$$

$$\mathbf{E}(t, \mathbf{x}) = -i \frac{\mu_0 I_0 c}{2\pi R} e^{i(kR - \omega t)} \frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \mathbf{e}_\theta \quad (5.172)$$

$$\langle S \rangle = \frac{\mu_0 I_0^2 c}{8\pi^2 R^2} \frac{\cos^2\left(\frac{\pi}{2} \cos \theta\right)}{\sin^2 \theta} \quad (5.173)$$

$$f(\theta) = \frac{\cos^2\left(\frac{\pi}{2} \cos \theta\right)}{\sin^2 \theta} \quad (5.174)$$

$$P = 2.44 \frac{\mu_0 I_0^2 c}{8\pi} \quad (5.175)$$

$$R_r = 2.44 \frac{\mu_0 c}{4\pi} = 73.2 \Omega \quad (5.176)$$

Prop. 44 天线阵

6 运动带电粒子与电磁场的相互作用

6.1 带电粒子的辐射

6.1.1 运动带电粒子的势

Def. 1 Liénard-Wiechert 势

$$\begin{cases} \mathbf{A} = \frac{q\mathbf{v}}{4\pi\epsilon_0 c^2 \left(r - \frac{\mathbf{v}}{c} \cdot \mathbf{r} \right)} \\ \phi = \frac{q}{4\pi\epsilon_0 \left(r - \frac{\mathbf{v}}{c} \cdot \mathbf{r} \right)} \end{cases} \quad (6.1)$$